

36. EATs

EATs_ET140s9

**Pro 6031T, Pro 6037H, Pro 6042H, Pro 6048H, Pro 6048 T, Pro 6046TT, Pro 6049TT,
Pro 6054TT, Pro 6055TT, Pro 6028T, Pro 6042HT, Pro 6035TM, Pro 8028XC, Pro
8028XM, Pro 8035XM, Pro 8055TT**

36. EATs

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36.1. Eats Removal And Installation

36.1.1 Catalytic Converter

36.1.1.1 Tools and Equipment:

- 13 mm combination spanner
- ½ inch drive ratchet with 13 mm socket bit
- Torque wrench (range 20 to 25 Nm)
- Personal protective gloves
- Safety goggles

36.1.1.2 Parts List / Requirements:

S.NO	Part Number	Description	Quantity
1	ID601934	Catalytic convertor	1
2	ID312534	Clamp	1
3	ID202261	U-bolt	2
4	ID381492	Gasket	1
5	MF430122	Nuts	4
6	MF140460	Mounting bolts	6

36.1.1.3 Work Time / Flat Rate Timing

- Estimated time: 45 minutes

36.1.1.4 Diagnostic Code Availability

- Not applicable for this procedure.

36.1.1.5 Pre-removal Procedure

- Ensure the vehicle is on level ground.
- Allow the exhaust system to cool completely.
- Disconnect the vehicle battery to prevent electrical hazards.
- Wear safety gloves and goggles.

36.1.1.6 Removal Procedure

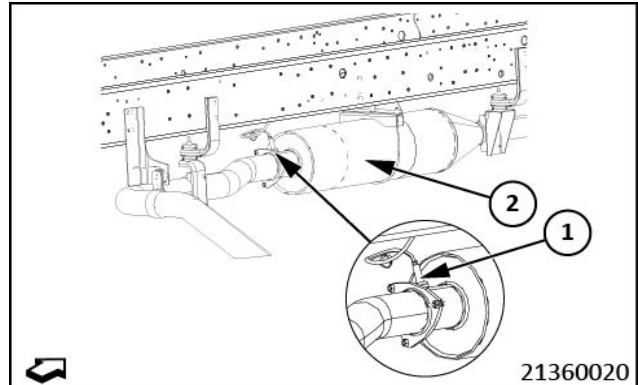
Warning

SCR catalytic convertor (1) is heavier than standard exhaust mufflers. Two or more people must hold the catalytic convertor (1) during removal to avoid personal injury.

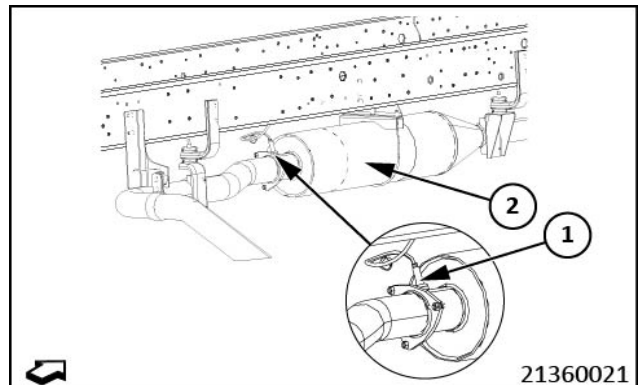
Warning

Exhaust components may be hot. Allow sufficient cooling time before beginning the procedure to prevent personal injury.

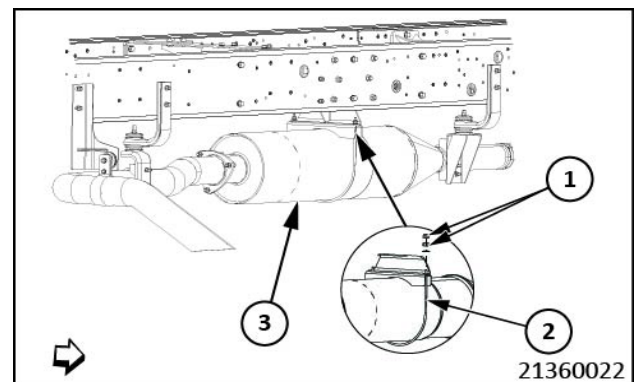
1. Disconnect the wiring harness connector (1) from the catalytic convertor (2).



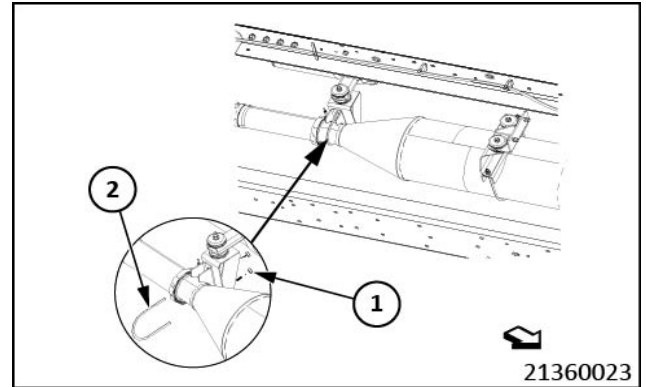
2. Loosen the clamp (1) located at the decomposition tube (2) side using a 13 mm spanner.



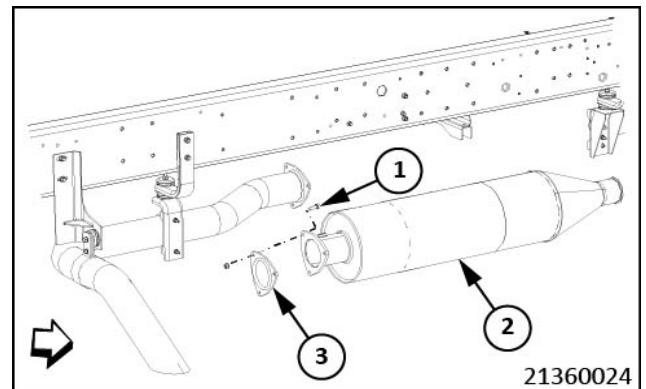
3. Remove the mounting nut (1) from the U-bolt (2) on the catalytic convertor (3) using a 13 mm socket and ratchet.



4. Loosen the mounting nut (1) on the other side of the U-bolt (2), then remove the U-bolt (2) and nut (1) entirely.

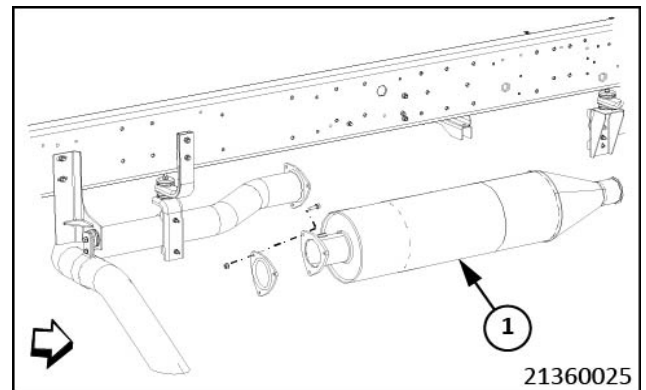


5. Remove the mounting bolt (1) of the catalytic convertor (2) from the tailpipe side and take out the catalytic convertor (2) along with the gasket (3).

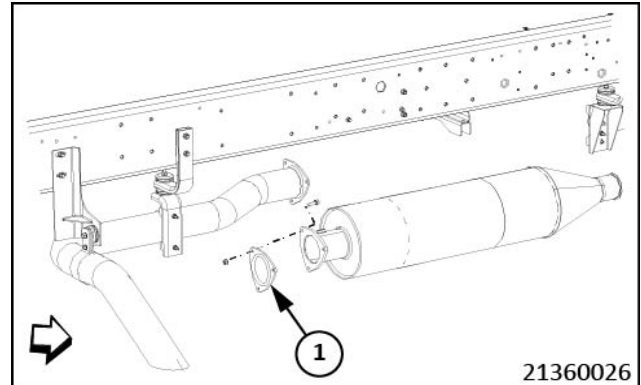


36.1.1.7 Inspection

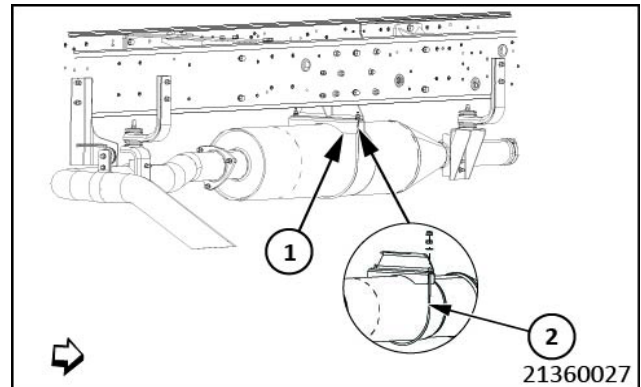
1. Check for cracks, corrosion, or carbon deposits on the catalytic convertor (2).



2. Ensure the gasket (3) is intact and not brittle or damaged.

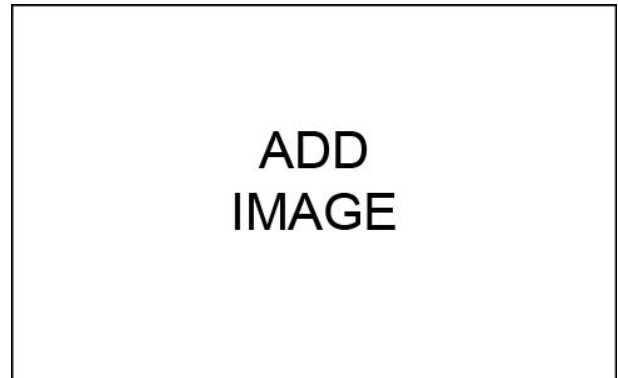


3. Verify the clamp (1), U-bolt (2), and mounting hardware are not worn or deformed.



36.1.1.8 Lubrication

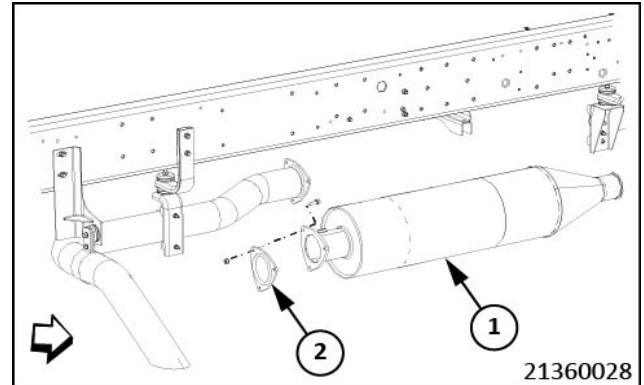
1. Apply anti-seize lubricant to the U-bolt (3) threads and mounting bolts (5) before installation.



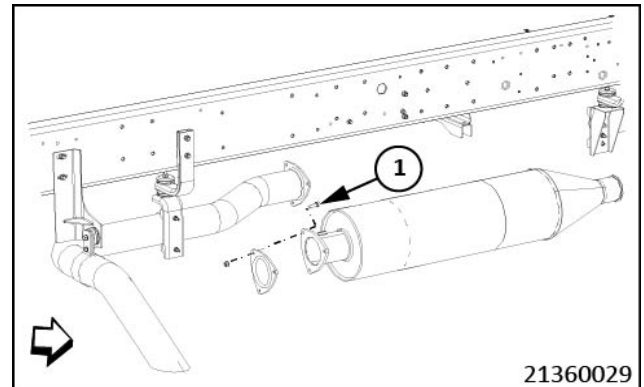
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36.1.1.9 Installation Procedure

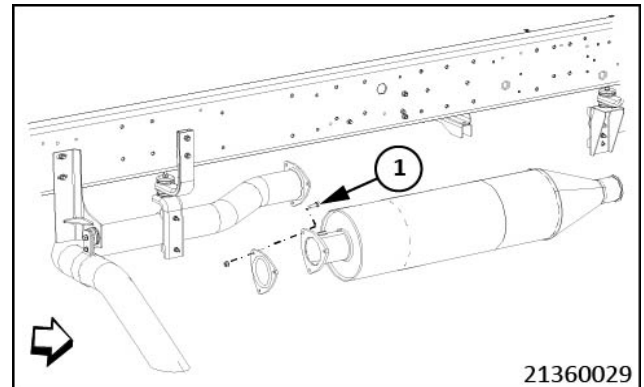
1. Align the catalytic convertor (2) with the tailpipe and position the gasket (3) correctly.



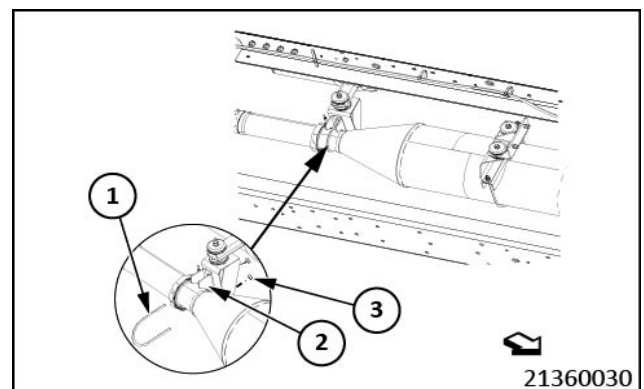
2. Tighten the mounting bolt (1) on the tailpipe side using a 13 mm socket and ratchet.



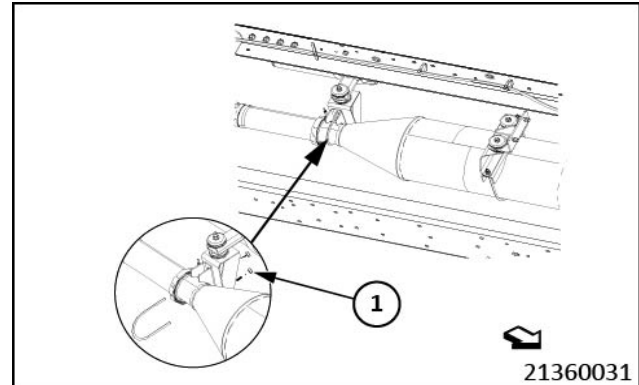
3. Torque the mounting bolt (1) between 20 to 25 Nm using a torque wrench.



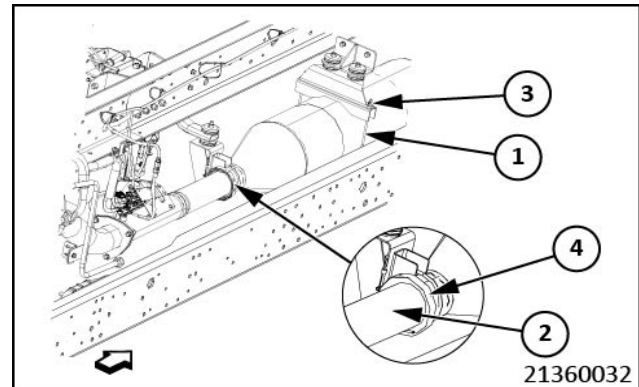
4. Position the U-bolt (1) with the catalytic converter support bracket (2) and tighten the mounting nuts (3) using a 13 mm spanner.



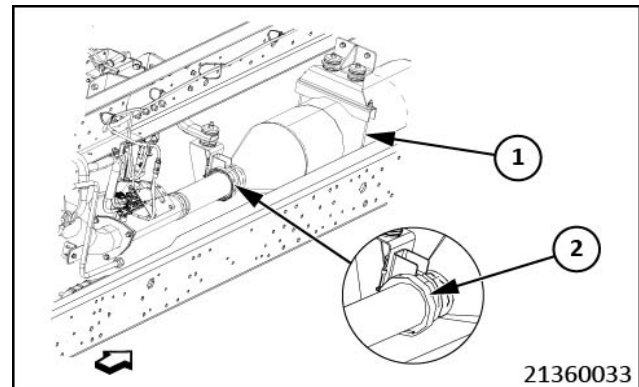
5. Torque both U-bolt nuts (1) between 20 to 25 Nm using a torque wrench.



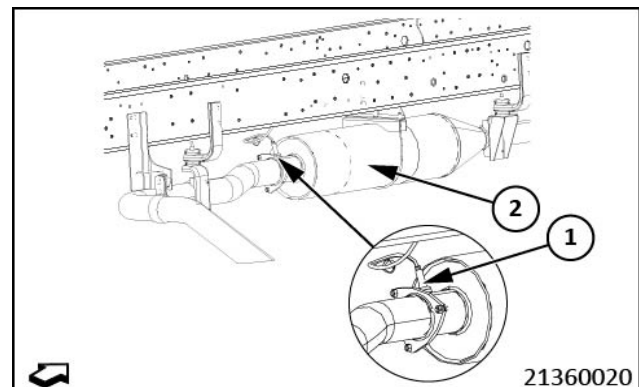
6. Attach the U-bolt (1) at the decomposition tube (2) side and tighten both mounting nuts (3). Then, secure the clamp (4) using a 13 mm spanner.



7. Torque all U-bolt (1) and clamp (2) bolts between 20 to 25 Nm using a torque wrench.

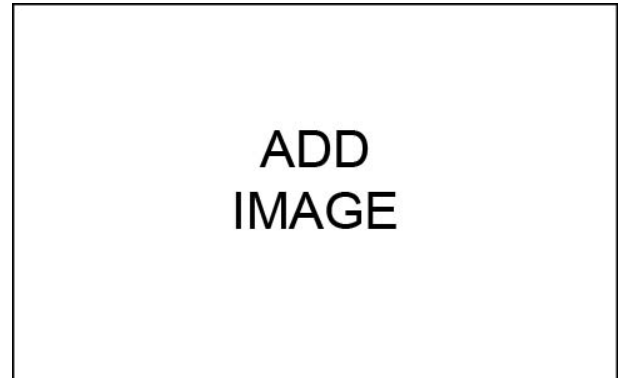


8. Reconnect the wiring harness (1) to the catalytic convertor (2).

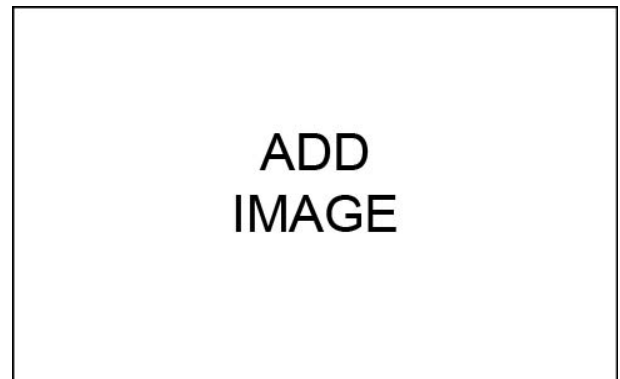
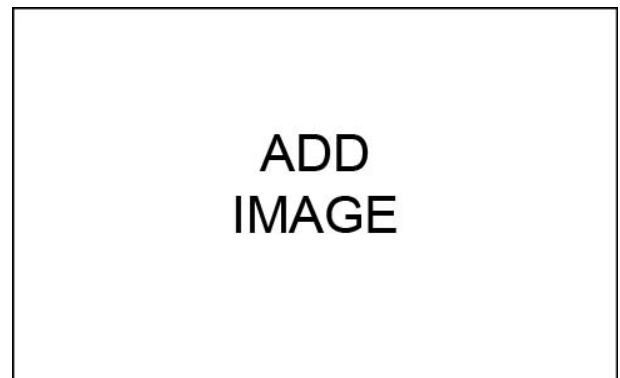


36.1.1.10 Alignment Check

1. Ensure the catalytic convertor (1) is aligned properly with the exhaust pipe and mounted without strain on joints or brackets.

**36.1.1.11 Final Testing**

1. Start the engine and inspect for any exhaust leaks.
2. Check that the catalytic convertor (1) and pipe joints are vibration-free and secure.

**36.1.1.12 Warranty Clause**

- Use only OEM-approved parts and torque specifications.
- Warranty may be void if procedure is not followed precisely using the recommended tools and specifications.

36.1.2 EATS ACM Harness**36.1.2.1 Tools and Equipment:**

- 10 mm spanner (for wing nut removal if tightened)

- Gloved hands (for handling connections and harness)

36.1.2.2 Parts List / Requirements:

S.NO	Part Number	Description	Quantity
1	NS (23-043-RC200100001D)	ACM harness (if replacement is required)	1
2		Washer (if damaged or rusted)	1
3	IF000327	Wing nut (if damaged or rusted)	1
4	IC419603	Top cover (UREA TANK DUST COVER ASSY)	1

36.1.2.3 Work Time / Flat Rate Timing

- Estimated time: 0.20 hours

36.1.2.4 Diagnostic Code Availability

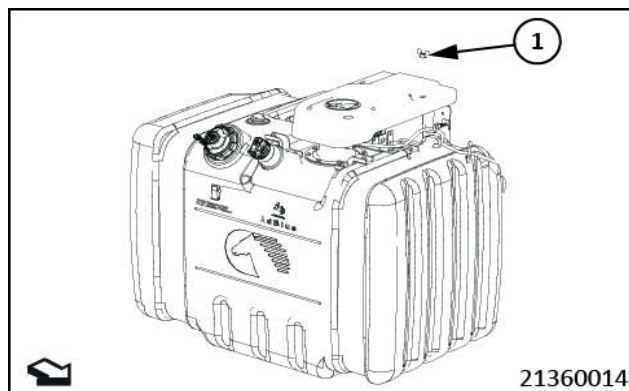
- Not applicable for mechanical removal and fitment task

36.1.2.5 Pre-removal Procedure

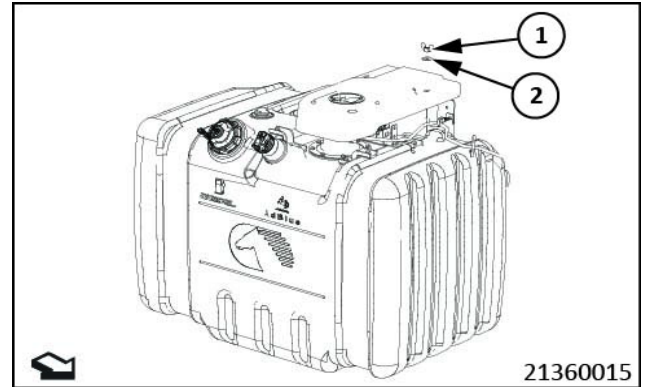
1. Park the vehicle on a level surface and switch off the ignition key.
2. Allow the urea tank and its components to cool completely before handling.

36.1.2.6 Removal Procedure

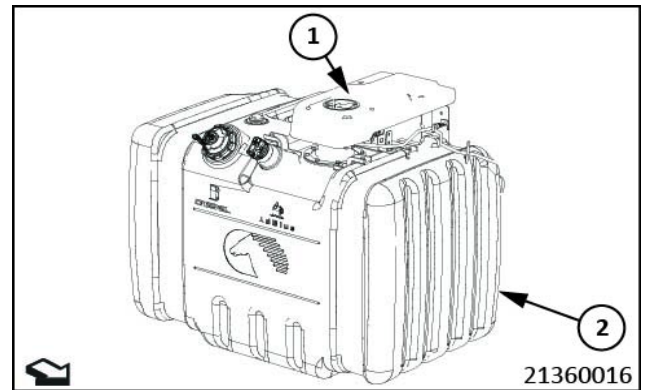
1. Use gloved hands or a 10 mm spanner to loosen the urea dust cover wing nut (1).



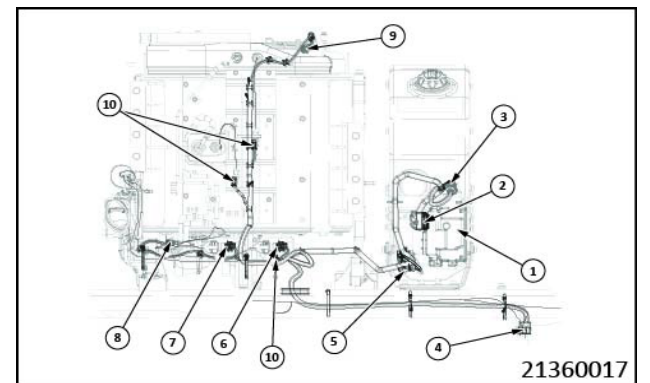
2. Remove the wing nut (1) along with the washer (2).



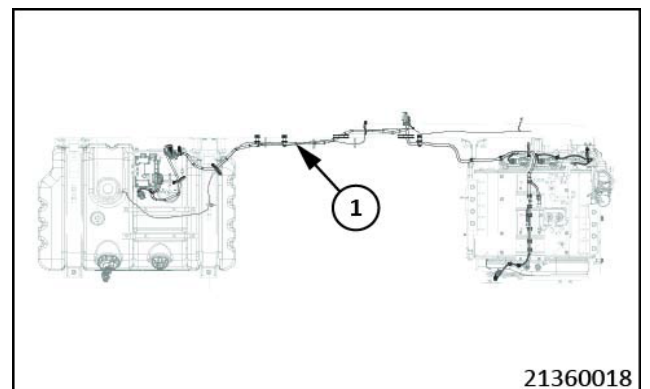
3. Lift and take out the top cover (1) from the urea tank (2).



4. Disconnect all ACM harness connectors carefully by hand.

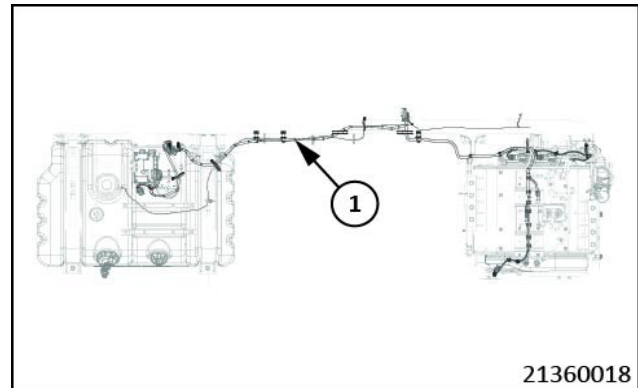


5. Gently pull and remove the ACM harness (1) from its installed position.

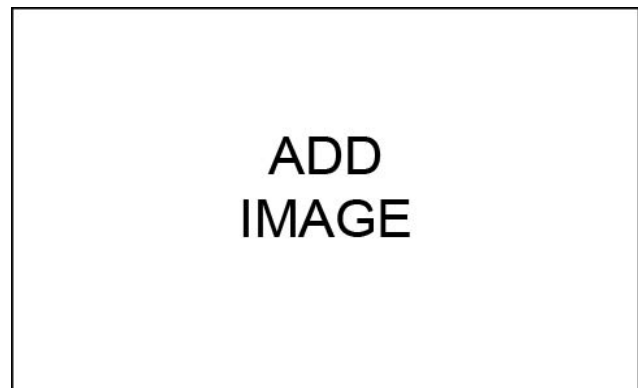


36.1.2.7 Inspection

1. Check the condition of the ACM harness (1) connectors for corrosion or damage.



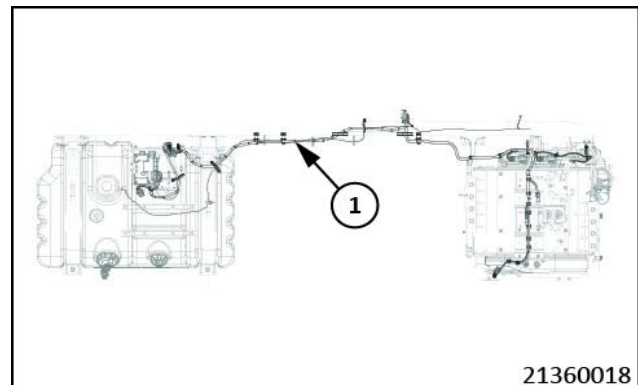
2. Ensure the dust cover, washer, and wing nut are free from cracks or wear.

**36.1.2.8 Lubrication**

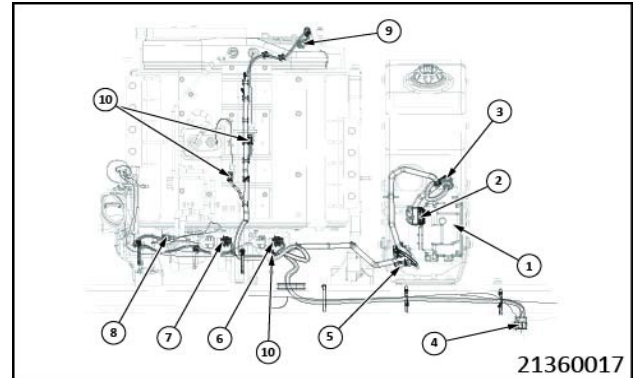
- Not applicable.

36.1.2.9 Installation Procedure

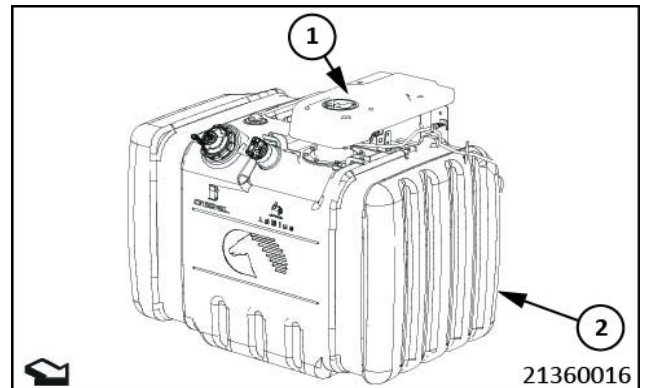
1. Position the ACM harness (1) correctly and route it along the original path.



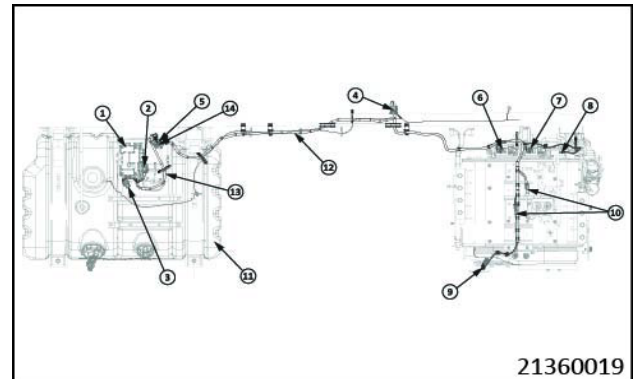
2. Connect all ACM harness connectors securely.



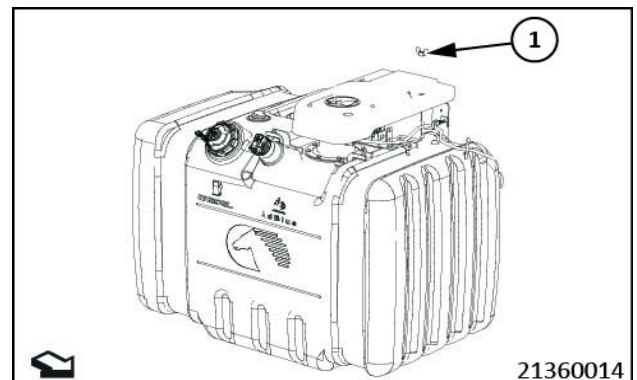
3. Place the top cover (1) back onto the urea tank (2).



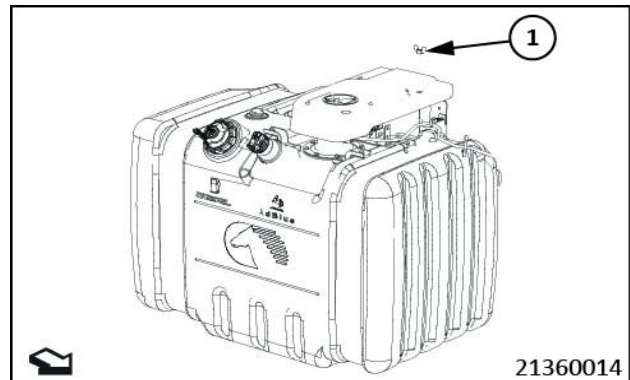
4. Install the washer (1) and wing nut (2) onto the dust cover stud.



5. Tighten the wing nut (1) securely by hand or using a 10 mm spanner.

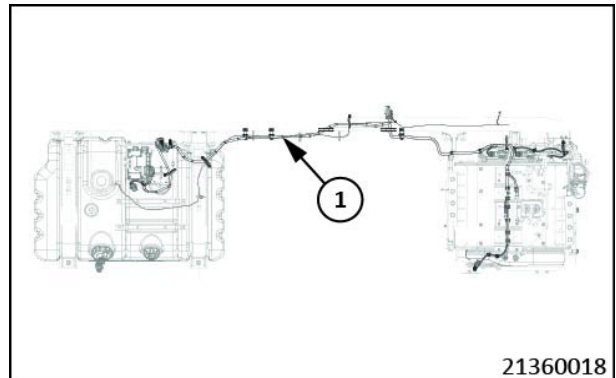


6. Torque the wing nut (1) to 5 to 7 Nm using a torque wrench with a 10 mm socket.

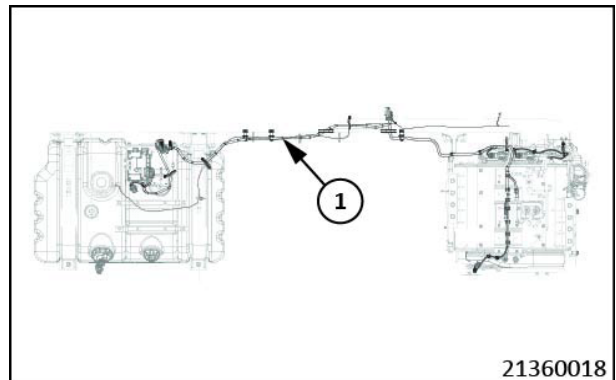


36.1.2.10 Alignment Check

1. Ensure the ACM harness (1) routing does not interfere with other vehicle components.



2. Confirm connectors are fully seated and locked.



36.1.2.11 Final Testing

1. Turn ON the ignition and observe the dashboard for any EATS-related warning messages.

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2. Use the diagnostic tool to verify ACM communication is active.

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36.1.2.12 Warranty Clause

Applicable warranty types:

- Parts Warranty
- Labour Warranty
- Standard Vehicle Warranty
- LCP Scheme (Labour, Consumables, Parts)

36.1.3 Eats Assembly

36.1.3.1 Tools and Equipment:

- 10 mm socket with ratchet
- 13 mm socket with ratchet
- Torque wrench (3/8 inch) with 10 mm and 13 mm sockets
- V-band clamp tightening tool
- Special tool trolley (for EATS muffler support)
- Anti-seize lubricant
- Protective gloves (for handling electrical connectors and components)

36.1.3.2 Parts List / Requirements:

S.NO	Part Number	Description	Quantity
1	ID312453	New V-band clamp (mandatory if previously reused)	1
2	MF450005	New washers bracket (if damaged)	7
3	MF140221	Bolts for muffler	13
4	ID370523	Muffler mounting hardware	1
5	Need to collect	Exhaust sealing gasket (if applicable)	1
6	ID366328	Bracket, Box Muffler	1
7	ID366900	ASSY FRONT EXHAUST PIPE	1

36.1.3.3 Work Time / Flat Rate Timing

- Estimated total time for removal and installation: 1.5 hours

36.1.3.4 Diagnostic Code Availability

- Fault codes may be generated if sensors are disconnected without prior power-down.
- Use the authorized diagnostic tool to scan and clear fault codes after installation.

36.1.3.5 Pre-removal Procedure

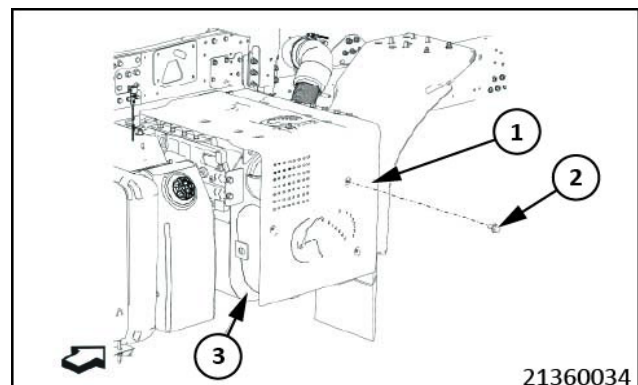
1. Park the vehicle on a level surface and engage the parking brake.
2. Turn off the ignition and wait for the EATS muffler to cool completely.
3. Wear safety gloves and protective eyewear.

Warning

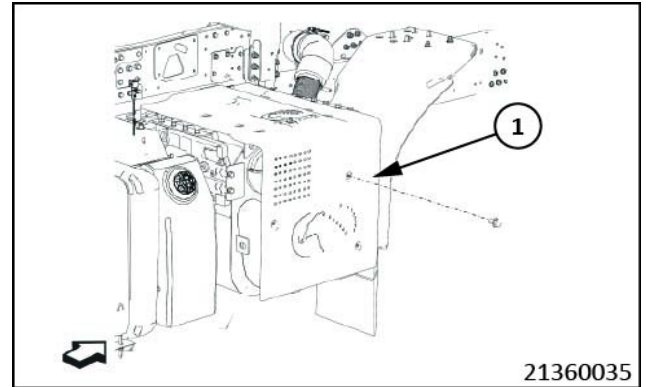
Do not touch the EATS muffler or exhaust system while hot. Wait until components are fully cooled to prevent personal injury.

36.1.3.6 Removal Procedure

1. Loosen and remove beauty shield (1) mounting bolts (2) using a 10 mm socket and ratchet.

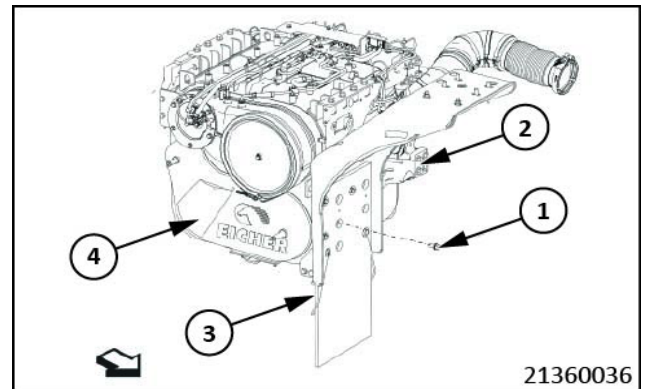


2. Remove the EATS muffler beauty shield (1).

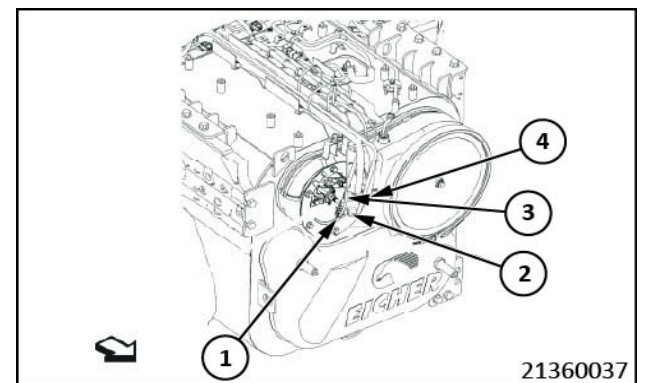


3. Loosen and remove the muffler mudguard bracket bolts (1) using a 10 mm socket and ratchet.

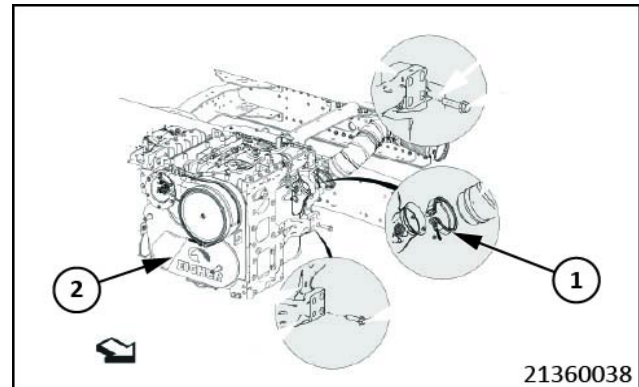
4. Detach the muffler mudguard bracket (2) and mudguard (3) from the EATS muffler (4).



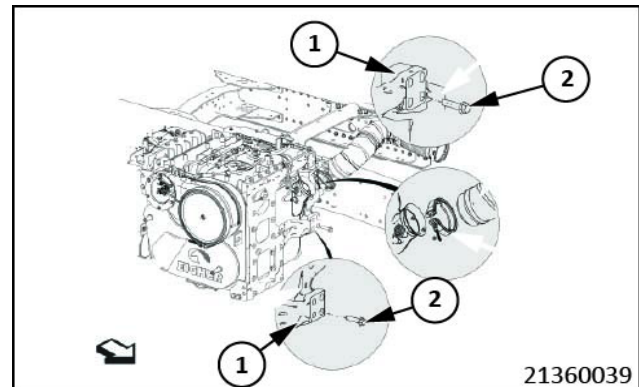
5. Disconnect the connectors from the urea dosing sensor(1), injection port (2), and coolant inlet (3) and outlet (4) ports using gloved hands.



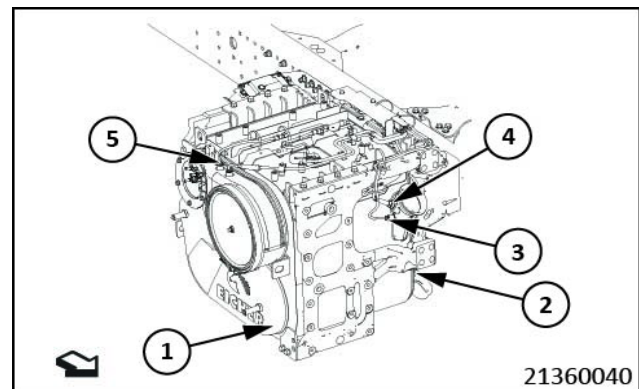
6. Loosen the V-band clamp (1) using the V-band clamp tool and disconnect the exhaust pipe from the muffler (2).



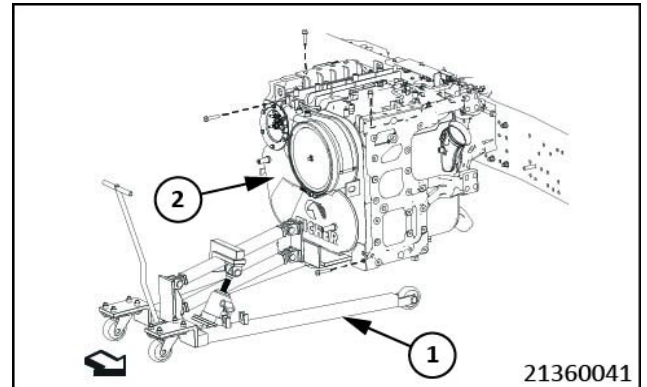
7. Remove the upper and lower muffler bracket (1) and mudguard bracket bolts (2) using a 13 mm socket and ratchet.



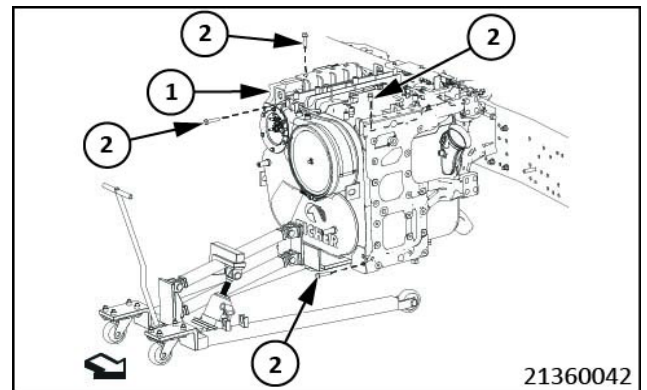
8. Disconnect all electrical connectors from the DOC (4), DPF, SCR temperature sensors (5), NOx sensors (3), and differential pressure sensor using gloved hands.



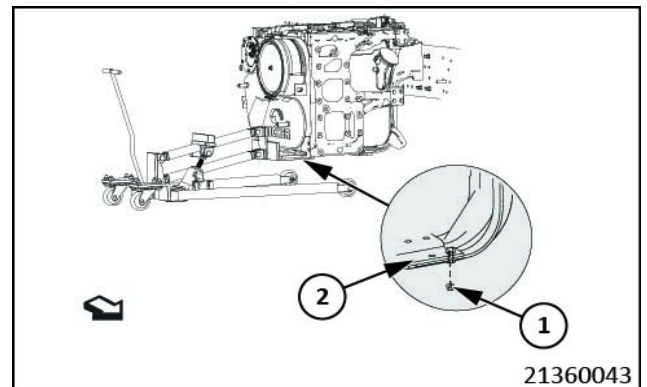
9. Place the special tool trolley (1) under the muffler (2) to support its weight.



10. Remove the LH upper muffler bracket (1) and mudguard bracket bolts (2) using a 10 mm socket and ratchet.



11. Remove the nut and washer (1) using a 13 mm socket and ratchet from the lower muffler bracket (2).

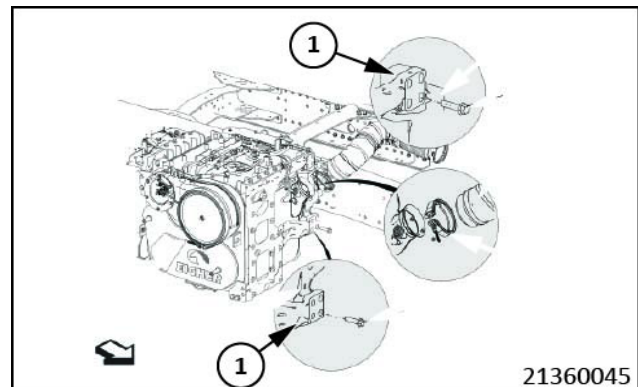
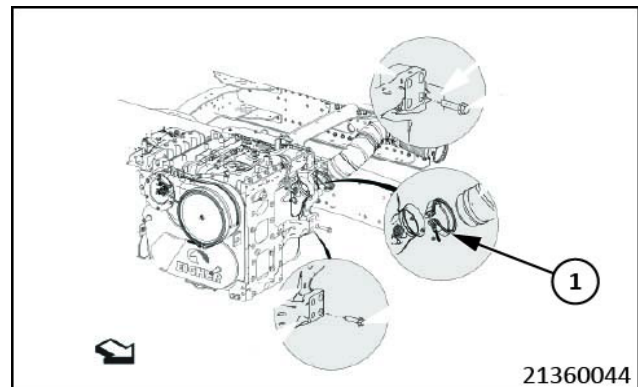


12. Carefully remove the complete EATS assembly using the trolley.

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36.1.3.7 Inspection

1. Check the V-band clamp (1) for cracks, deformation, or wear.
2. Inspect mounting brackets (1) and bushings for damage.



3. Ensure all sensor connectors are free from corrosion and damage.

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4. Clean all mating surfaces of soot, rust, and debris.

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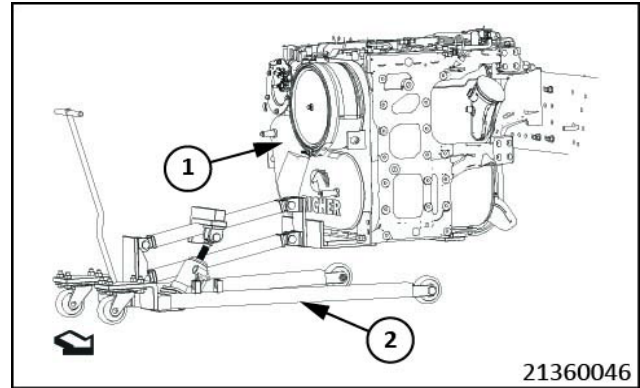
36.1.3.8 Lubrication

1. Apply anti-seize lubricant to all mounting bolts, V-band clamp threads, and sensor threads before installation.

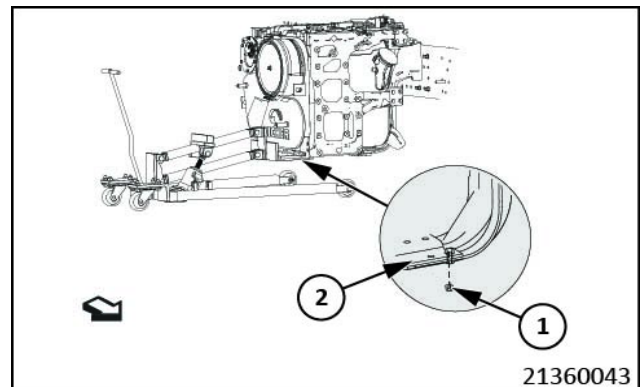
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36.1.3.9 Installation Procedure

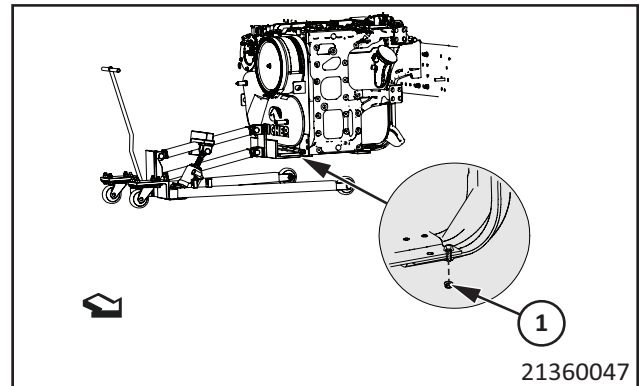
1. Position the EATS muffler (1) on the special tool trolley (2) and align it to the vehicle chassis.



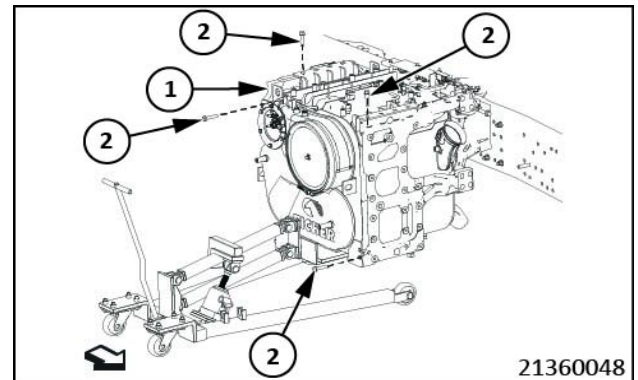
2. Install the nut and washer (1) to the lower muffler bracket (2) and tighten using a 13 mm socket and ratchet.



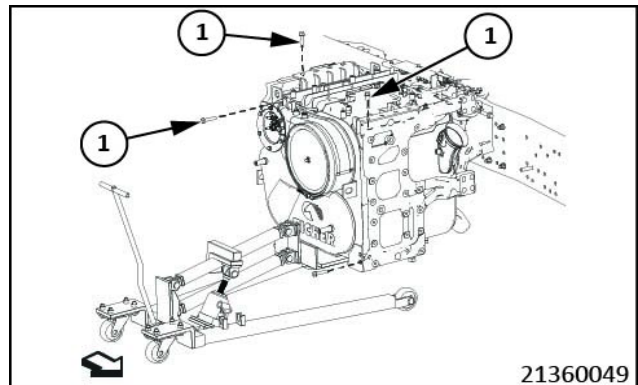
3. Torque the lower muffler bracket nut (1) to 22 to 25 Nm using a torque wrench with a 13 mm socket.



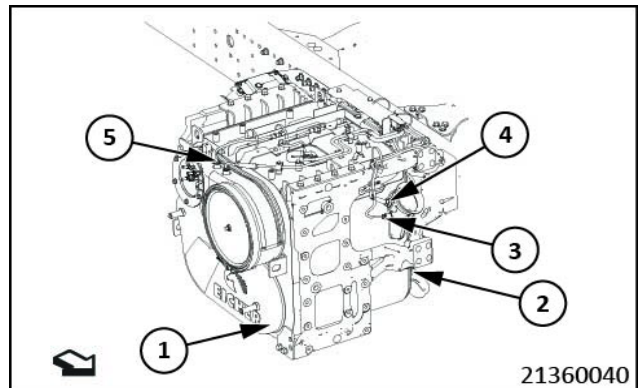
4. Install the LH upper muffler bracket (1) and mudguard bracket bolts (2) and tighten using a 10 mm socket and ratchet.



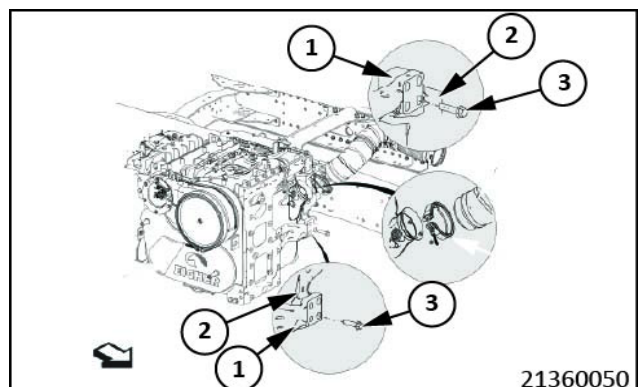
5. Torque the LH upper bracket bolts(1) to 8 to 11 Nm using a torque wrench with a 10 mm socket.



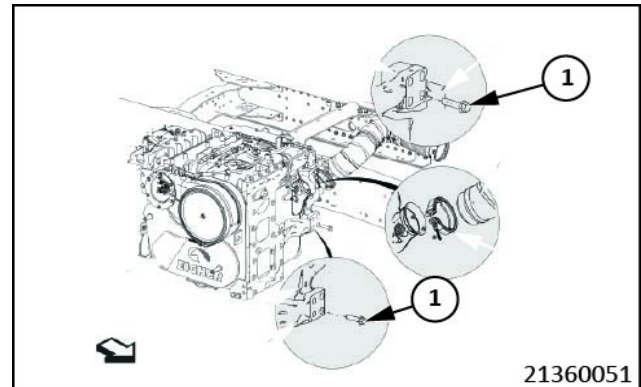
6. Reconnect all electrical connectors to the DOC, DPF, SCR, NOx, and differential pressure sensors using gloved hands.



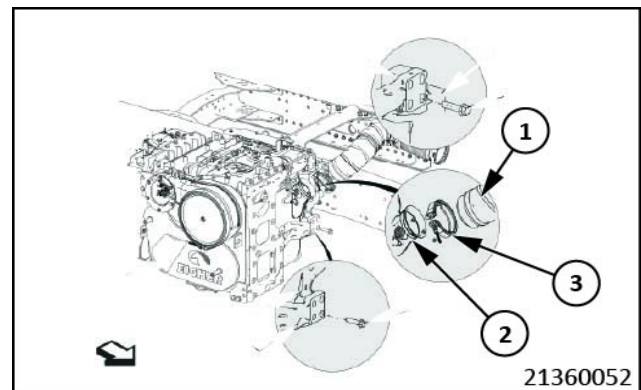
7. Install the upper and lower muffler mudguard bracket (1) to the muffler bracket frame (2) using a bolt (3) and tighten using 10 mm socket and ratchet.



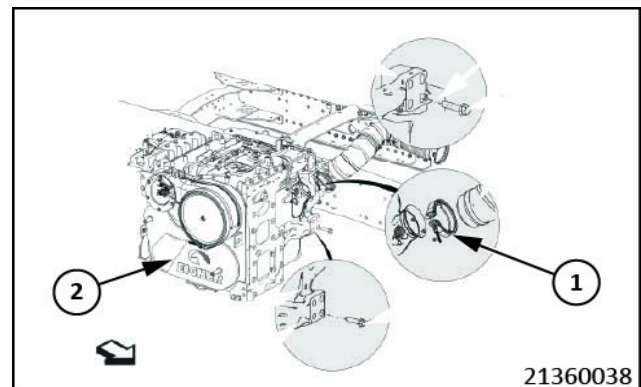
8. Torque the mudguard bracket bolts (1) to 8 to 11 Nm using a torque wrench with a 10 mm socket.



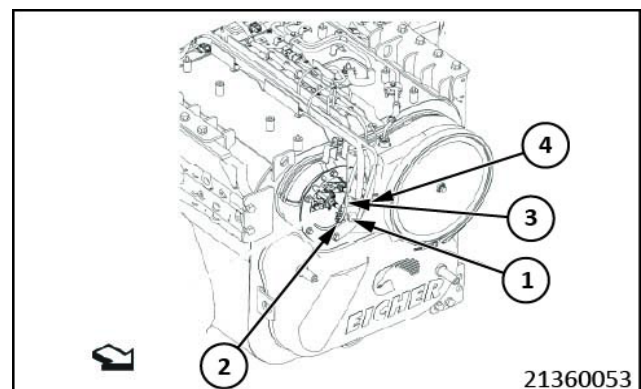
9. Align the exhaust pipe (1) with the muffler outlet (2) and secure it with the V-band clamp (3) using the V-band clamp tool.



10. Torque the V-band clamp (1) to 5 to 7 Nm using a torque wrench.

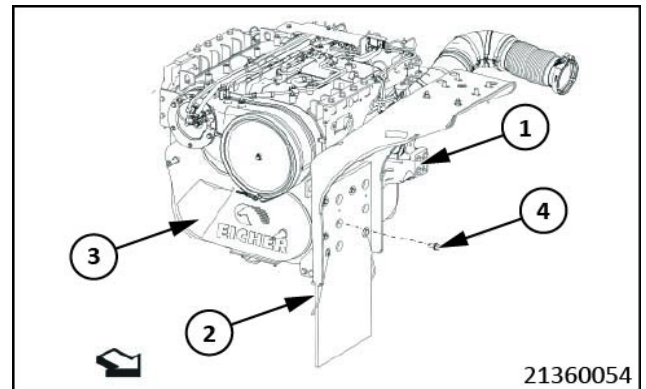


11. Connect the urea injection port (1), dosing sensor (2), and coolant lines (3) (4) using gloved hands.

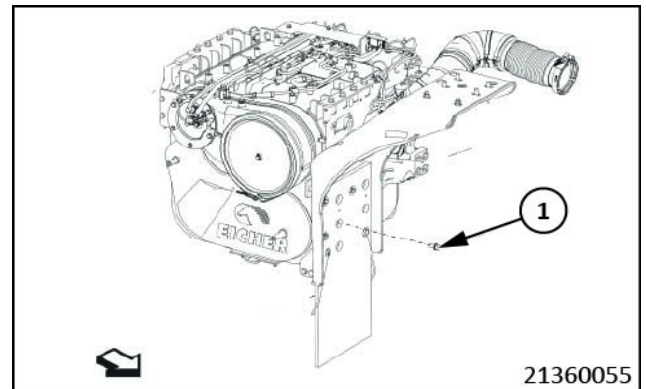


12. Attach the muffler mudguard bracket (1) and mudguard (2) from the EATS muffler (3).

13. Install the muffler mudguard bracket bolts (4) and tighten using a 10 mm socket and ratchet.

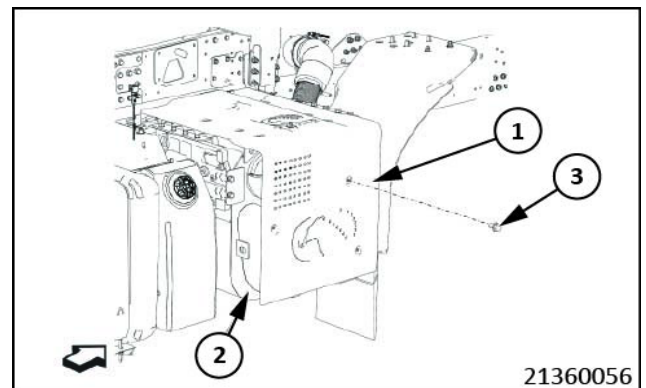


14. Torque the muffler mudguard bracket bolts (1) to 8 to 11 Nm using a torque wrench with a 10 mm socket.

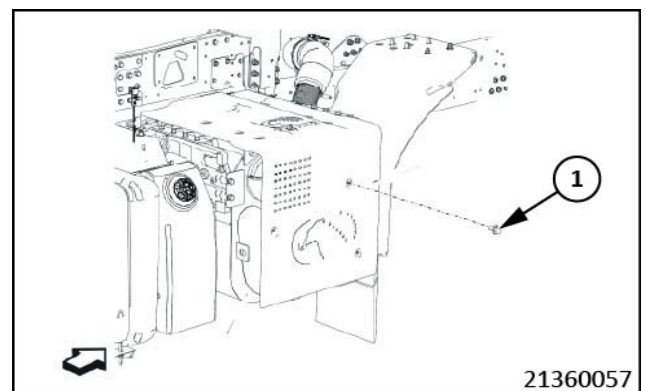


15. Place the EATS beauty shield (1) over the EATS muffler (2).

16. Install the mounting bolts (3) to the EATS beauty shield (1) and tighten using a 10 mm socket and ratchet.



17. Torque the beauty shield bolts (1) to 8 to 11 Nm using a torque wrench with a 10 mm socket.



36.1.3.10 Alignment Check

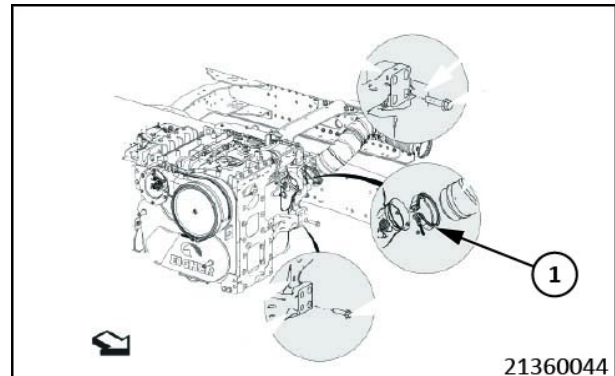
1. Ensure the EATS muffler is properly aligned with the exhaust flange.

ADD
IMAGE

2. Confirm there is no stress or misalignment in the bracket assembly or sensor routing.

ADD
IMAGE

3. The V-band clamp (1) should be seated evenly without any gap.

**36.1.3.11 Final Testing**

1. Start the engine and allow it to idle.

ADD
IMAGE

2. Use a diagnostic tool to check for any active fault codes.

ADD
IMAGE

3. Monitor live sensor data to confirm proper functioning.

ADD
IMAGE

4. Inspect for any signs of leakage or abnormal noise.

ADD
IMAGE

5. Conduct a short road test to verify EATS operational status.

ADD
IMAGE

36.1.3.12 Warranty Clause

Applicable warranty coverage under the following categories:

- Standard Vehicle Warranty
- Aggregate Warranty
- Parts Warranty
- Labour Warranty
- Consumables Warranty
- Extended / Driveline Warranty
- LCP Scheme (Labour, Consumables, Parts)
- AMC (Annual Maintenance Contract)
- Eicher Promise (Uptime / Service Assurance)
- Goodwill Warranty (Subject to approval)

36.1.4 Urea Quality Level Sensor (UQLS)

36.1.4.1 Tools and Equipment:

- 10 mm socket and ratchet
- Torque wrench (3/8 inch) with 10 mm socket
- Flathead screwdriver
- Clean dry cloth
- Approved urea-compatible lubricant
- Protective gloves

36.1.4.2 Parts List / Requirements:

S.NO	Part Number	Description	Quantity
1	ID374268	UQLS sensor assembly (if replacement required)	1
2	ID374268	UQLS Kit (includes new filter, O-ring, and retaining clip)	1
3	IC419111	Sensor gasket (replace if damaged)	1
4	IC334540	UQLS sensor clamp	1
5	IC419113	Sensor lid and top cover (reuse if undamaged)	
6	IC419547	Fasteners and washers (if damaged)	

36.1.4.3 Work Time / Flat Rate Timing

- Estimated time for removal and installation: 0.7 hours

36.1.4.4 Diagnostic Code Availability

- Fault codes may appear if the sensor is removed with the ignition ON.
- Use a diagnostic tool to verify and clear codes after installation.

36.1.4.5 Pre-removal Procedure

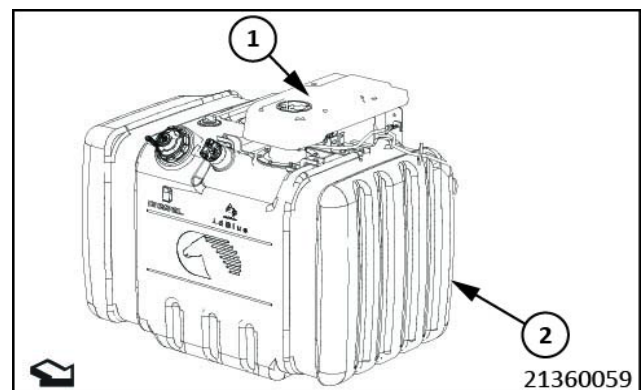
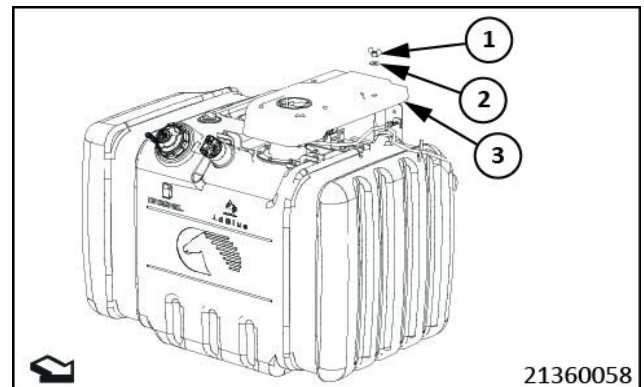
1. Park the vehicle on a level surface.
2. Switch off the ignition.
3. Allow the urea system to cool before handling.
4. Wear protective gloves and safety eyewear.

Caution

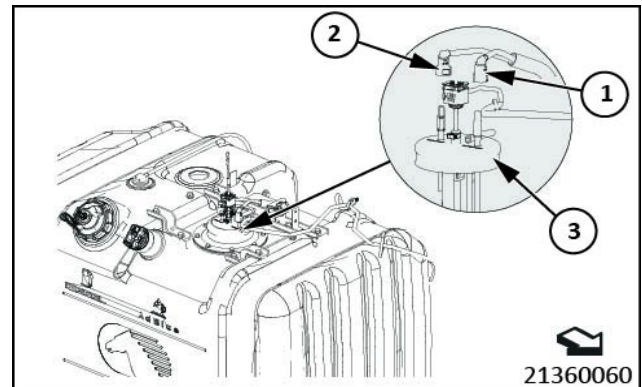
Do not drop the sensor. If the sensor is dropped, do not reuse it. Internal damage may not be visible but can affect performance.

36.1.4.6 Removal Procedure

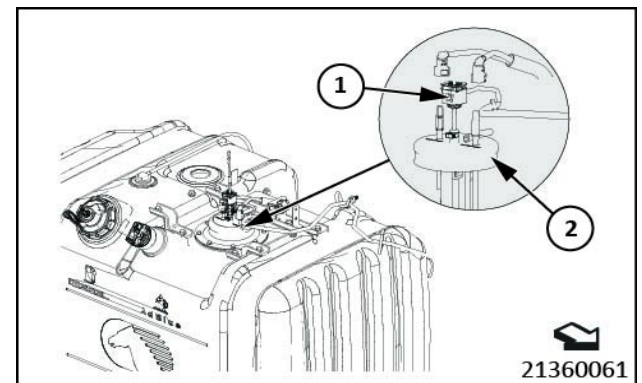
1. Loosen and remove the wing nut (1) and washer (2) from the top cover (3) of the urea tank using a 10 mm socket and ratchet.
2. Remove the top cover (1) from the urea tank (2) using gloved hands.



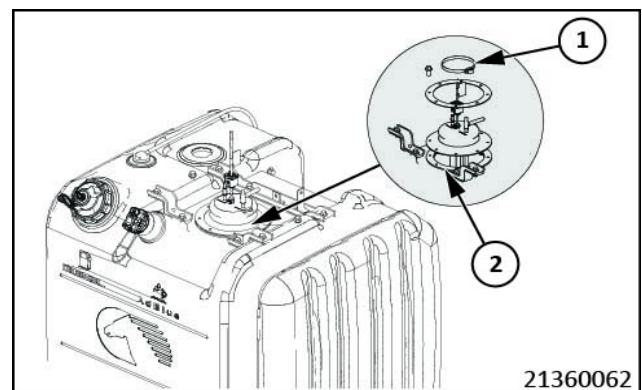
3. Disconnect the urea outlet (1) and backflow connections (2) from the UQLS ports (3) using gloved hands.



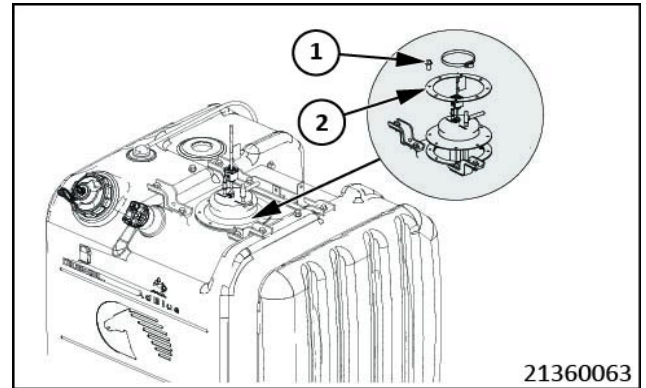
4. Disconnect the electrical connector (1) from the UQLS sensor (2) using gloved hands.



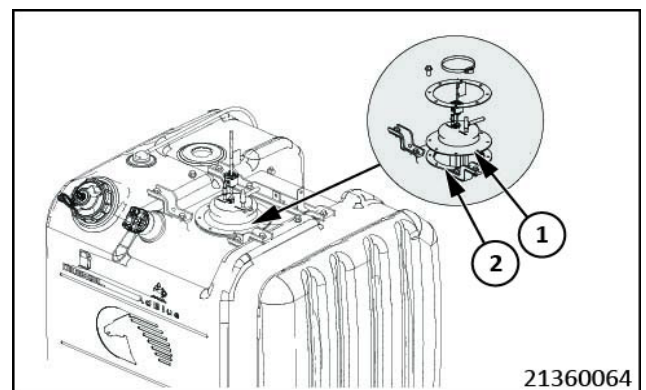
5. Remove the clamp (1) securing the UQLS sensor (2) using a flathead screwdriver.



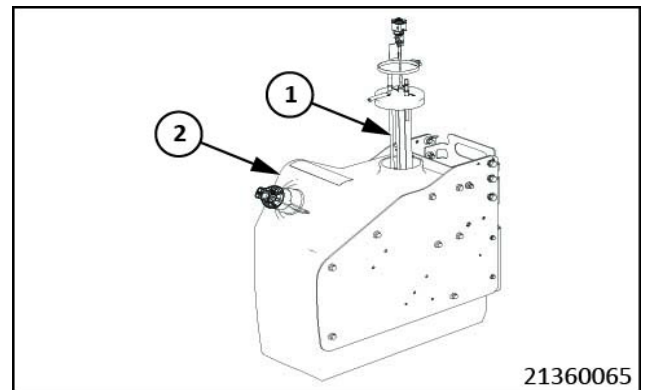
6. Loosen and remove the bolt (1) from the top metal plate (2) using a 10 mm socket and ratchet.



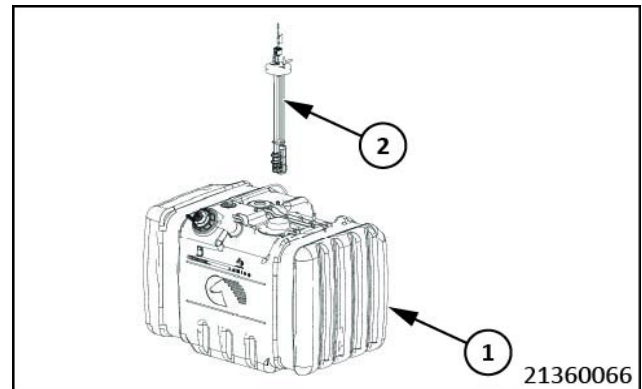
7. Remove the sensor lid (1) along with the gasket (2).



8. Carefully take out the urea quality level sensor (1) assembly from the AdBlue tank (2) using gloved hands.

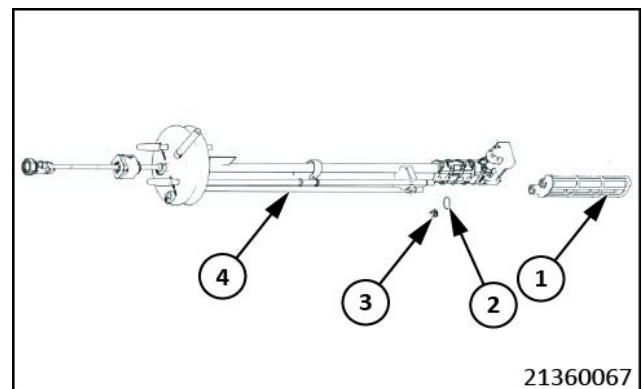


9. For tandem fuel tank(1) vehicles, rotate the sensor (2) from 0 to 180 degrees before removal to prevent interference.

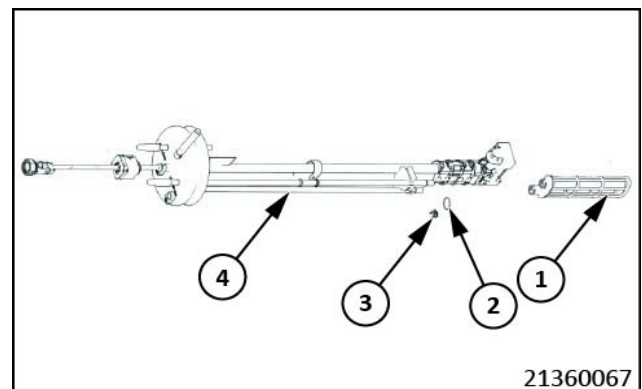


36.1.4.7 Inspection

1. Inspect the sensor for cracks, corrosion, or contamination.



2. Check the filter (1) O-ring (1) and clip (3) for wear or deformation. Replace if necessary.



3. Ensure connectors are clean, dry, and undamaged.

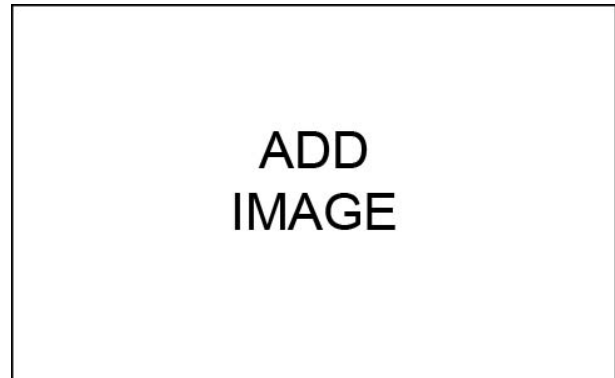
ADD
IMAGE

4. Clean any dirt or crystallized urea from the mounting surfaces.

ADD
IMAGE

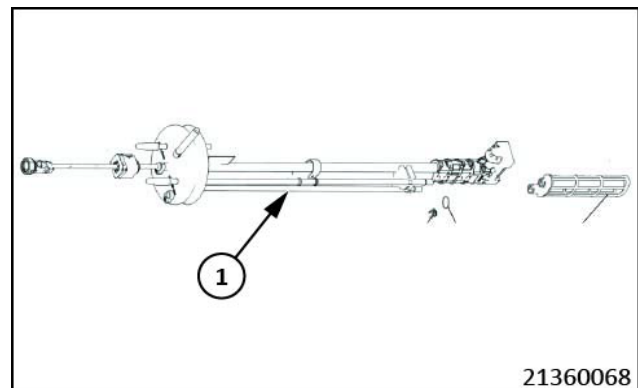
36.1.4.8 Lubrication

1. Apply approved urea-compatible lubricant to the new O-ring before installation.

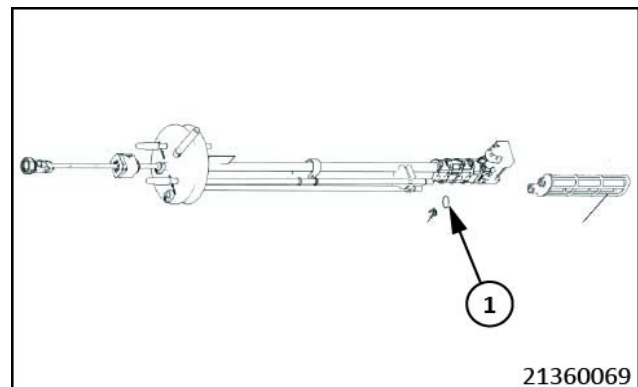


36.1.4.9 Installation Procedure

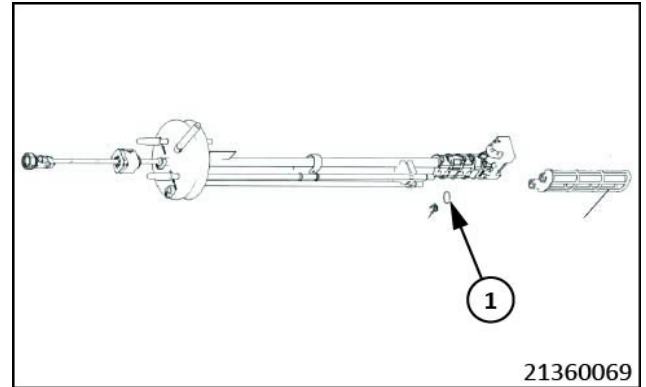
1. Clean the sensor (1) housing and mating surfaces using a clean dry cloth.



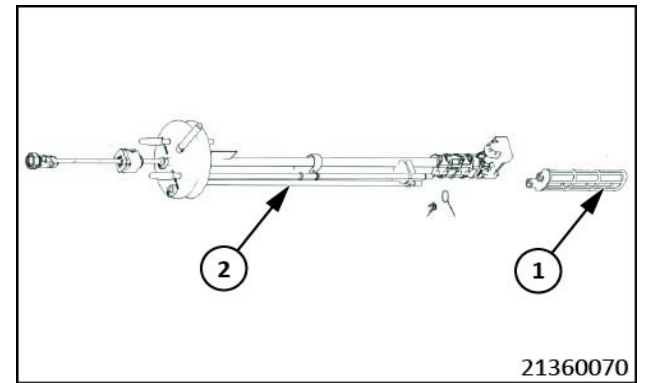
2. Insert the new O-ring (1) into the groove on the sensor port using gloved hands.



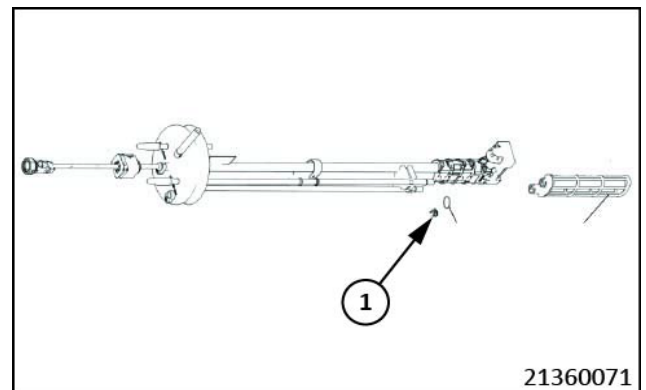
3. Apply the approved urea-compatible lubricant to the O-ring (1).



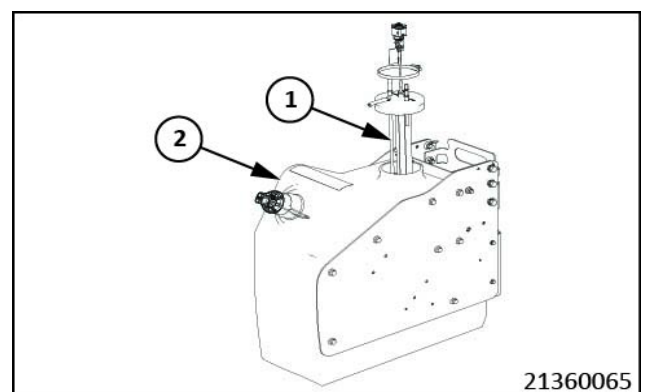
4. Insert the new filter element (1) into the UQLS (2) housing until fully seated.



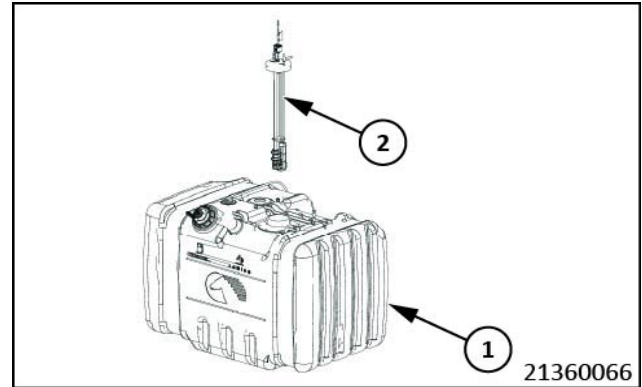
5. Install the new retaining clip (1) from the UQLS kit. Use a flathead screwdriver to ensure it locks properly.



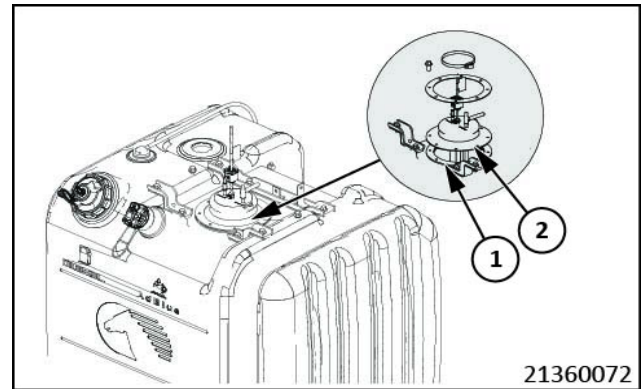
6. Carefully position the UQLS sensor (1) assembly into the AdBlue tank (2) using gloved hands.



7. For tandem fuel tank (1) vehicles, rotate the sensor (2) from 180 to 0 degrees during insertion.

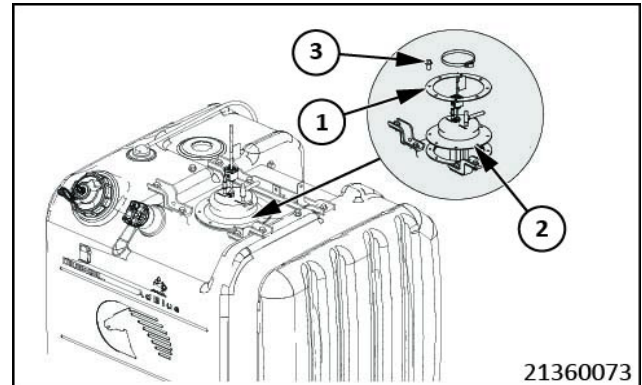


8. Align and place the gasket (1) and sensor lid (2) onto the tank opening.

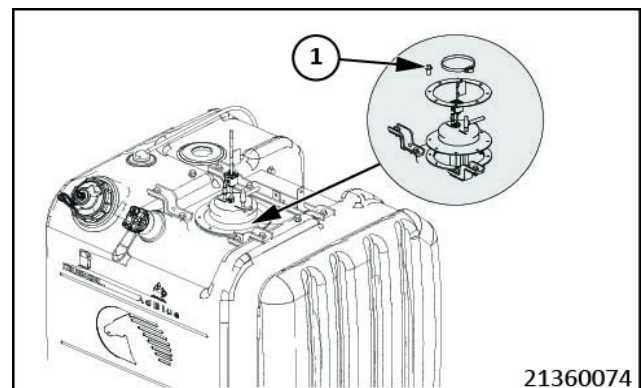


9. Place the top metal plate (1) onto the sensor lid (2).

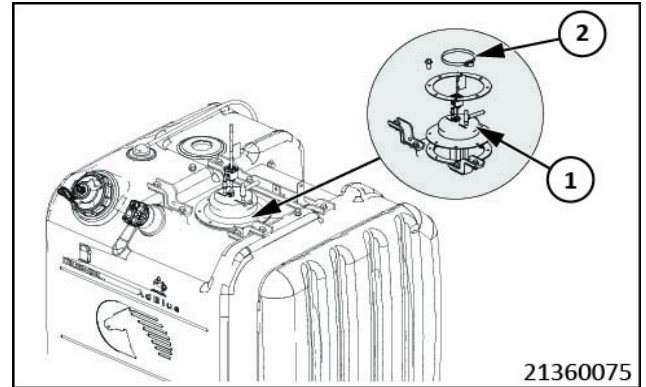
10. Insert and tighten the bolt (3) using a 10 mm socket and ratchet.



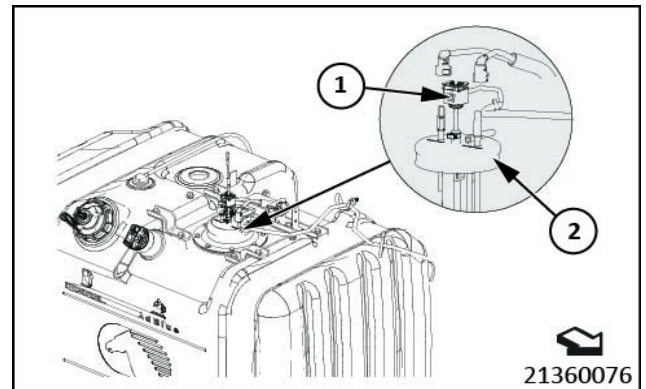
11. Torque the metal plate bolt (1) to 8 to 11 Nm using a torque wrench with a 10 mm socket.



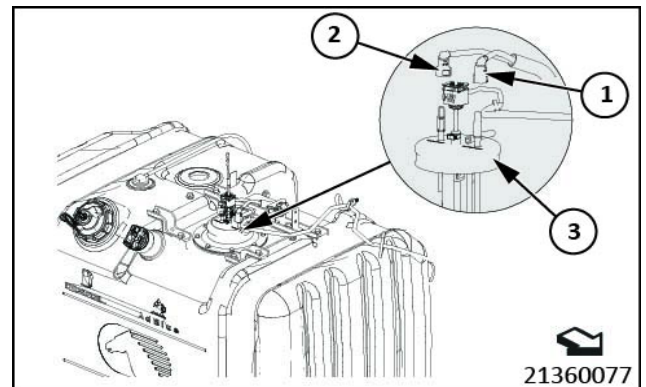
12. Secure the UQLS sensor (1) with the clamp (2) using a flathead screwdriver.



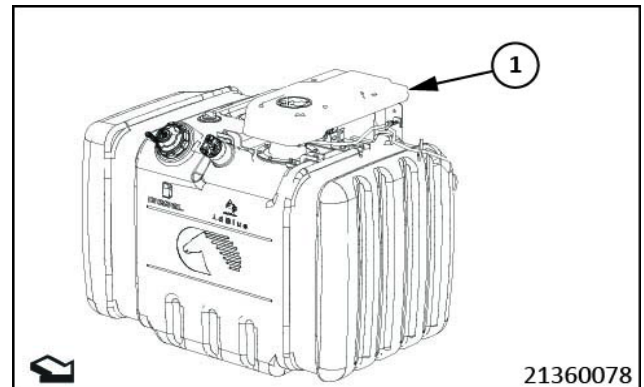
13. Connect the electrical connector (1) to the sensor (2) using gloved hands.



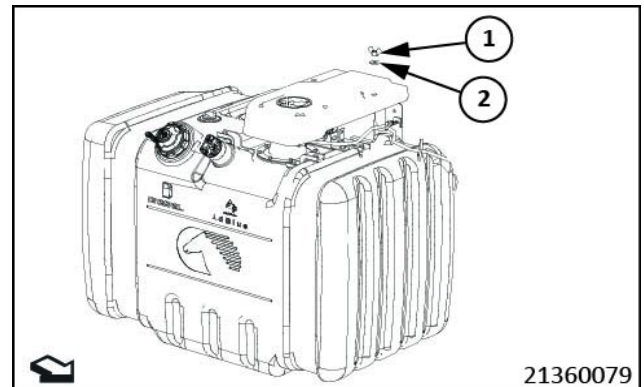
14. Reconnect the urea outlet (1) and backflow (2) lines to the sensor ports (3) using gloved hands.



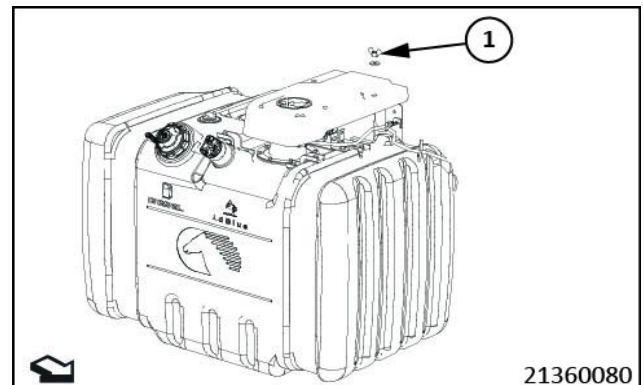
15. Place the urea tank top cover(1).



16. Fit and tighten the wing nut (1) along with the washer (2) using a 10 mm socket and ratchet.



17. Torque the wing nut (1) to 5 to 7 Nm using a torque wrench with a 10 mm socket.



36.1.4.10 Alignment Check

1. Ensure the sensor is seated flat and centered in the tank port.

ADD
IMAGE

2. Verify that hose and connector routing is correct and stress-free.

ADD
IMAGE

3. Confirm the top cover is flush and secured to the tank body.

ADD
IMAGE

36.1.4.11 Final Testing

1. Start the engine and monitor the system.

ADD
IMAGE

2. Check for any fault codes using a diagnostic tool.

ADD
IMAGE

3. Verify that urea level and quality readings are displayed correctly.

ADD
IMAGE

4. Inspect for any leaks or loose connections at sensor and hose ports.

ADD
IMAGE

36.1.4.12 Warranty Clause

The following warranty types apply:

- Standard Vehicle Warranty
- Parts Warranty
- Labour Warranty
- Consumables Warranty
- Aggregate Warranty
- Goodwill Warranty (subject to approval)
- Extended / Driveline Warranty
- LCP Scheme (Labour, Consumables, Parts)

- AMC (Annual Maintenance Contract)
- Uptime / Eicher Promise

36.1.5 Measurement of AdBlue Concentration

36.1.5.1 Tools and Equipment:

- Digital refractometer (Tool No. ST125498)
- Clean dry cloth
- Clean water (for post-test cleaning)
- Gloves (chemical resistant, for fluid handling)

36.1.5.2 Parts List / Requirements:

S.NO	Part Number	Description	Quantity
1	Need to check	Fresh AdBlue sample collected from the vehicle tank or test source	1
2	Need to check	Sampling container (clean and free from contaminants)	1

36.1.5.3 Work Time / Flat Rate Timing

- Estimated time for testing: 5 minutes per sample

36.1.5.4 Diagnostic Code Availability

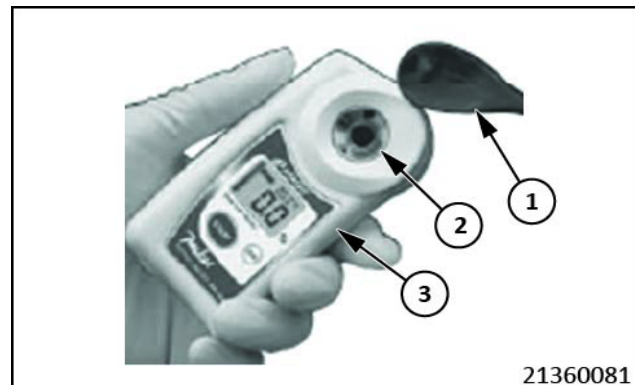
- Not applicable. This is a manual measurement step.

36.1.5.5 Pre-Measurement Procedure

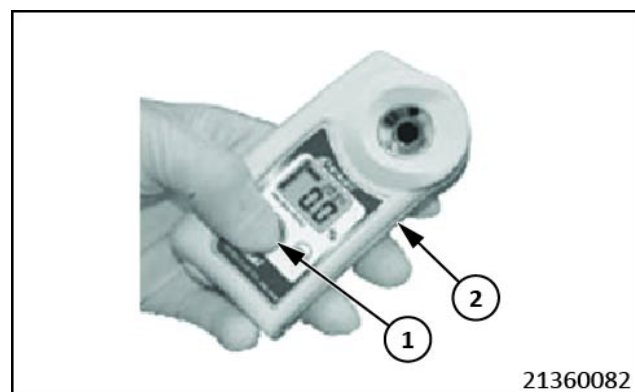
1. Park the vehicle on a level surface.
2. Switch off the ignition.
3. Wait for the system to cool before collecting the AdBlue sample.
4. Wear gloves and safety eyewear.
5. Ensure refractometer lens surface (prism) is clean and dry.

36.1.5.6 Measurement Procedure

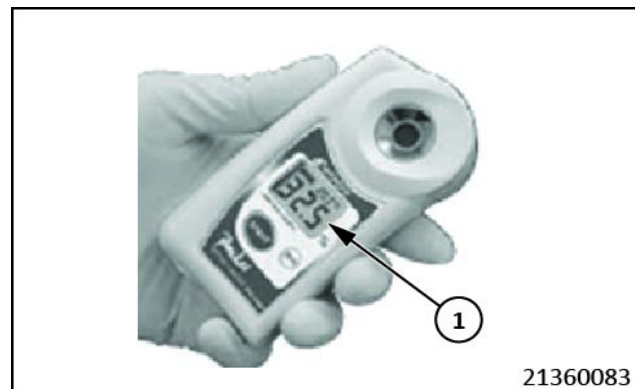
1. Use ST125498 Refractometer tool to measure the AdBlue concentration level
2. Place a small quantity of AdBlue sample (1) directly on the prism surface (2) of the refractometer (3) using a dropper or clean pipette.



3. Press the Start key (1) on the refractometer (2).



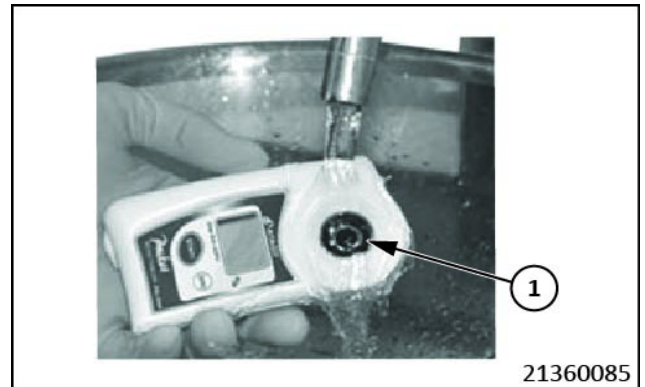
4. Wait approximately 3 seconds for the digital display (1) to show the measured concentration.



5. Note the displayed value. AdBlue should typically read 31.8% to 33.2% urea concentration.



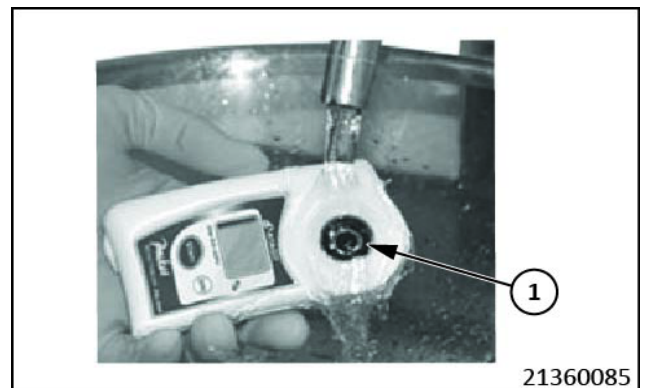
6. Remove the sample from the prism surface (1).



7. Clean the prism (1) using clean water and wipe dry with a clean, soft cloth.

Note

Clean the refractometer with water after every test to avoid contamination and ensure accurate results.



36.1.5.7 Final Testing and Interpretation

1. If the AdBlue concentration falls outside the acceptable range (31.8%–33.2%), replace the AdBlue with fresh fluid from an approved source.



2. Do not reuse tested samples.



36.1.5.8 Warranty Clause

Applicable warranty coverage if found defective or off-spec:

- Consumables Warranty (AdBlue quality-related claims)
- Parts Warranty (if UQLS or injector affected due to poor AdBlue quality)
- Goodwill Warranty (subject to approval in specific cases)

36.1.6 Vertical Exhaust – Removal and Installation Procedure

36.1.6.1 Tools and Equipment:

- 13 mm socket and ratchet
- 17 mm socket and ratchet
- V-band clamp tool
- Torque wrench (3/8 inch) with 13 mm and 17 mm sockets
- Gloved hands

36.1.6.2 Parts List / Requirements:

S.NO	Part Number	Description	Quantity
1	ID332677	New V-band clamp (if reused previously)	1
2	Need to check	Gasket for V-band joints	1
3	MF140229	Exhaust bracket bolts (if damaged)	1
4	MF434104	nuts	1
5	Need to check	Vertical exhaust stay	
6	MF450155	Washers (replace if damaged)	

36.1.6.3 Work Time / Flat Rate Timing

- Estimated time for complete removal and installation: 1.2 hours

36.1.6.4 Diagnostic Code Availability

- Not applicable for this mechanical operation.

36.1.6.5 Pre-removal Procedure

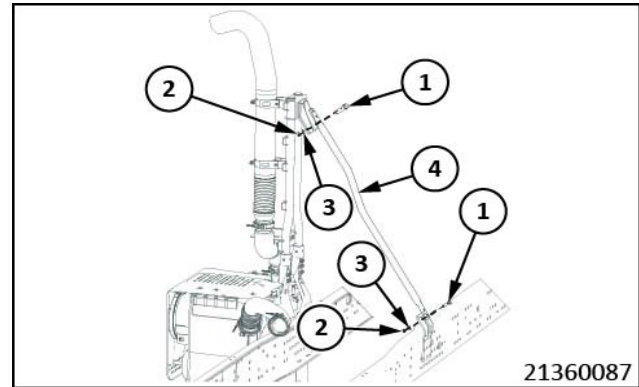
1. Park the vehicle on a level surface and apply the parking brake.
2. Switch off the ignition and allow the exhaust system to cool down completely.
3. Wear gloved hands and safety eyewear before beginning the procedure.
4. Wear protective gloves and safety eyewear.

Caution

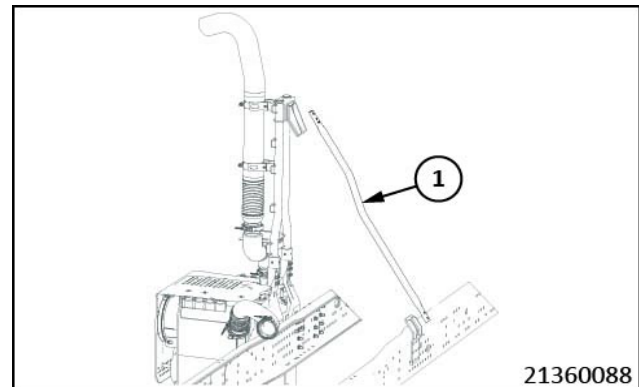
Do not start PTO work until the transmission is fully drained to prevent oil spillage.

36.1.6.6 Removal Procedure

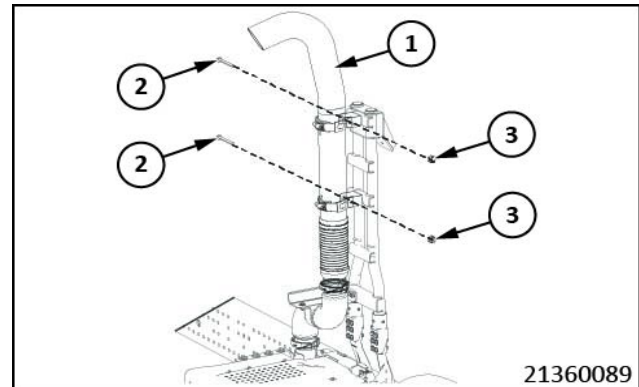
1. Loosen and remove the upper and lower mounting bolts (1) and nuts (2) along with washers (3) from the vertical exhaust stay (4) using a 13 mm socket and ratchet.



2. Remove the vertical exhaust stay (1) from the vehicle using gloved hands.

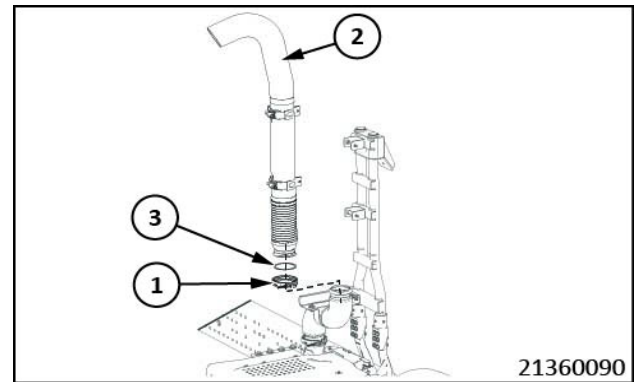


3. Loosen and remove the mounting of vertical exhaust (1) clamp bolt (2) and nut (3) using a 13 mm socket and ratchet.

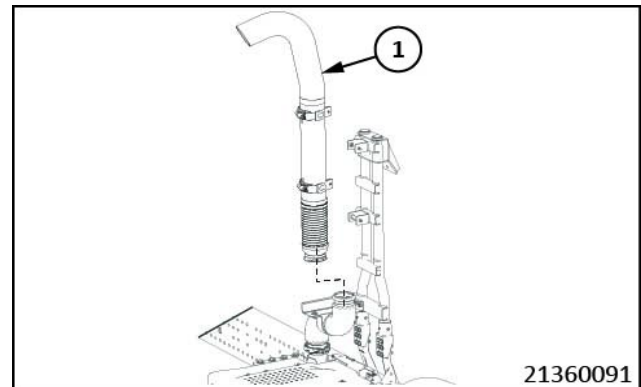


4. Loosen and remove the V-band clamp (1) that connects the vertical exhaust tail pipe (2) to the centre pipe exhaust using a V-band clamp tool.

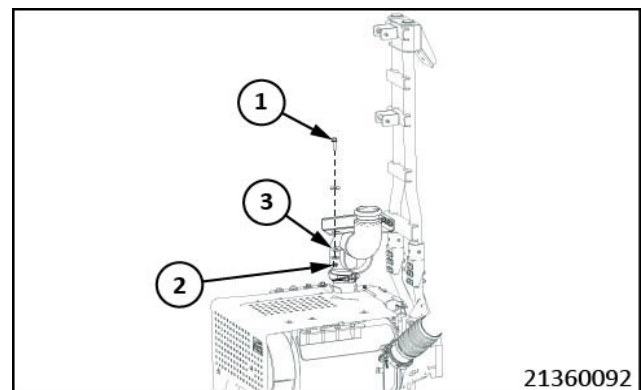
5. Remove the gasket (3) and take out the vertical exhaust tail pipe (2) using gloved hands.



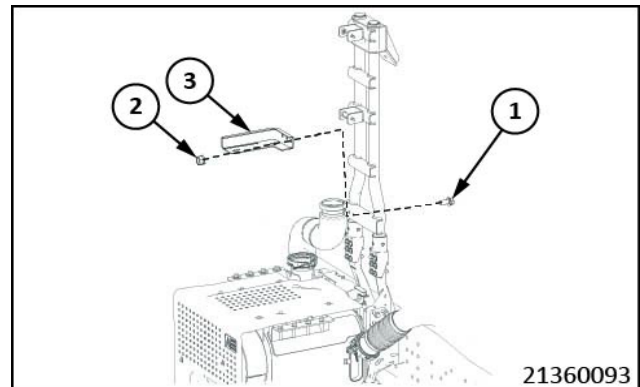
6. Take out the vertical exhaust tail pipe (1) using gloved hands.



7. Loosen and remove the clamp bolt (1) and nut (2) from the center pipe exhaust (3) using a 17 mm socket and ratchet.

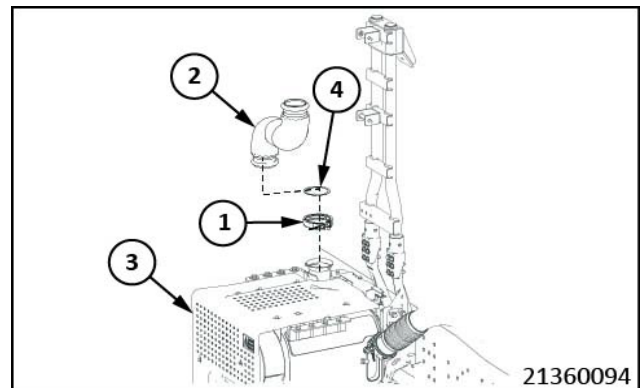


8. Loosen and remove the bracket mounting bolt (1) and nut (2), then take out the exhaust bracket (3) from the vertical exhaust using a 13 mm socket and ratchet.

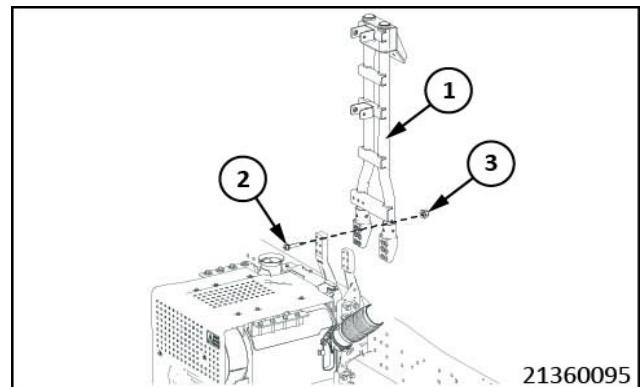


9. Loosen and remove the V-band clamp (1) connecting the center pipe (2) to the muffler (3).

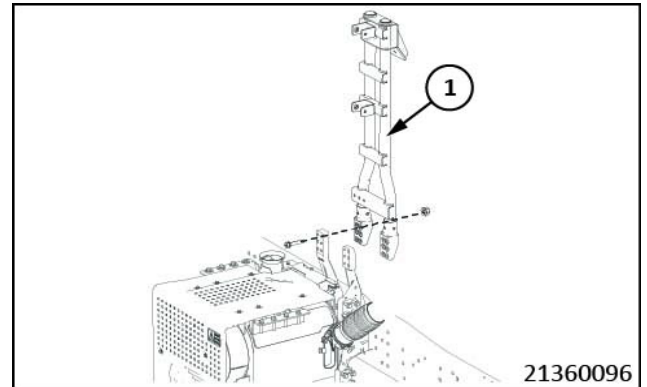
10. Remove the centre pipe exhaust (2) along with the clamp (1) and gasket (4) using a V-band clamp tool.



11. Loosen and remove the vertical exhaust bracket (1) mounting bolt (2) and nut (3) using a 13 mm socket and ratchet.

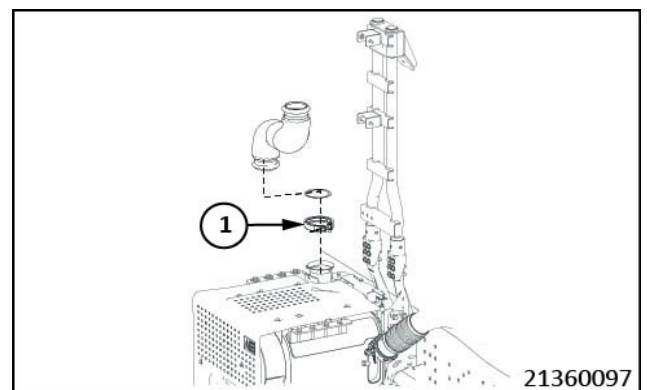


12. Take out the vertical exhaust bracket (1) using gloved hands.

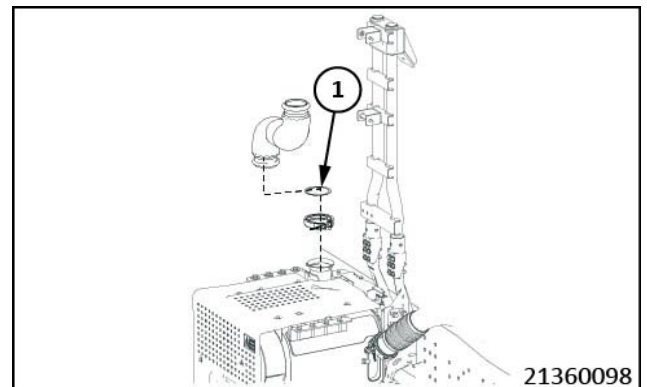


36.1.6.7 Inspection

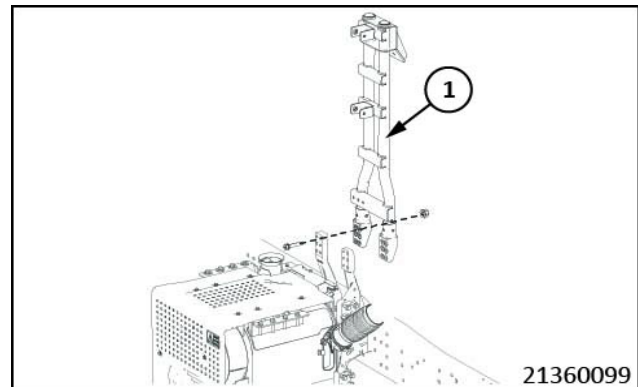
1. Inspect all V-band clamps (1) for wear or damage.



2. Check all gaskets (1) for carbon leaks and cracks.

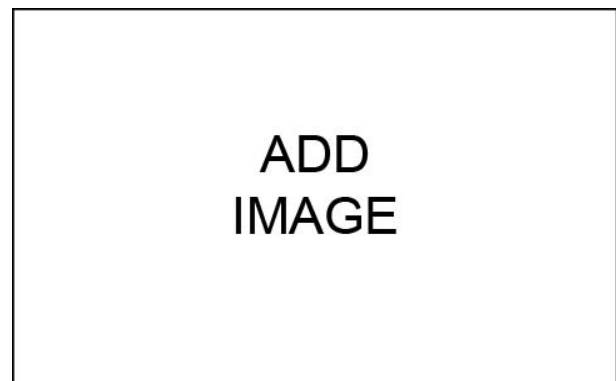


3. Ensure the bracket (1) threads and stay are not damaged or bent.
4. Replace any worn or damaged fasteners.



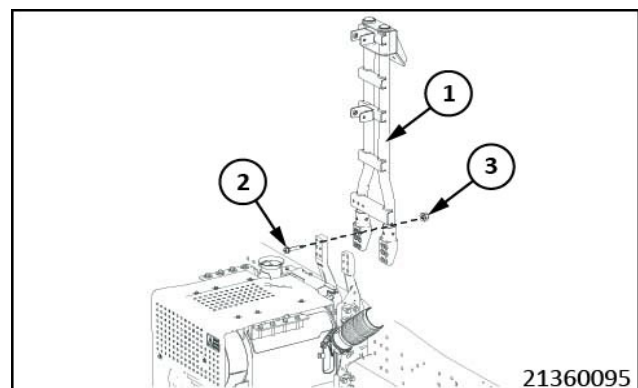
36.1.6.8 Lubrication

1. Apply anti-seize lubricant on all exhaust bolts and clamp threads before installation.

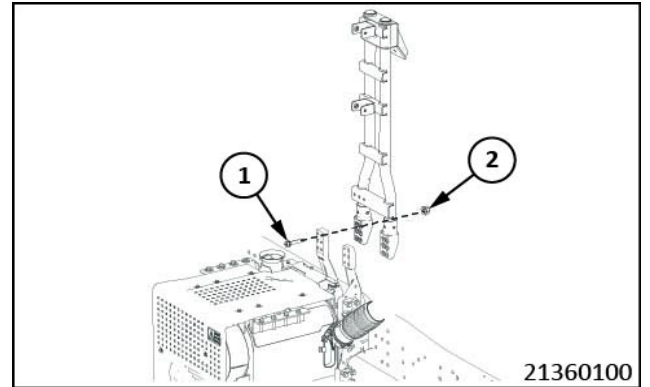


36.1.6.9 Installation Procedure

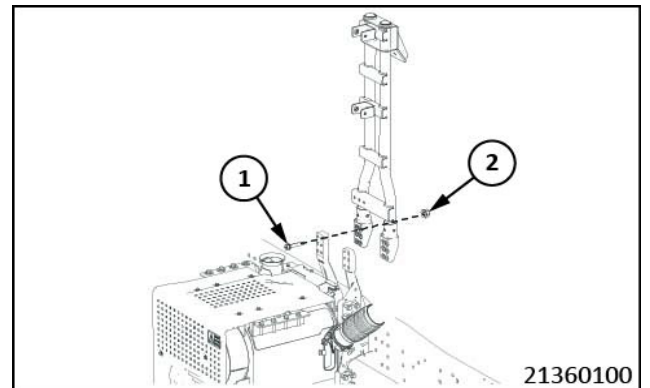
1. Position the vertical exhaust bracket (1) and insert the mounting bolt (2) and nut (3).



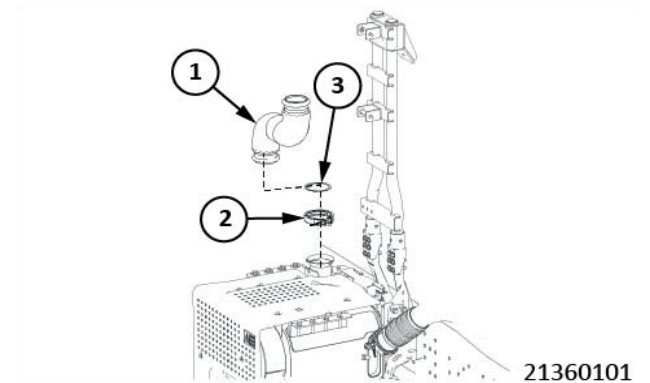
2. Tighten the mounting bolt (1) and nut (2) using a 13 mm socket and ratchet.



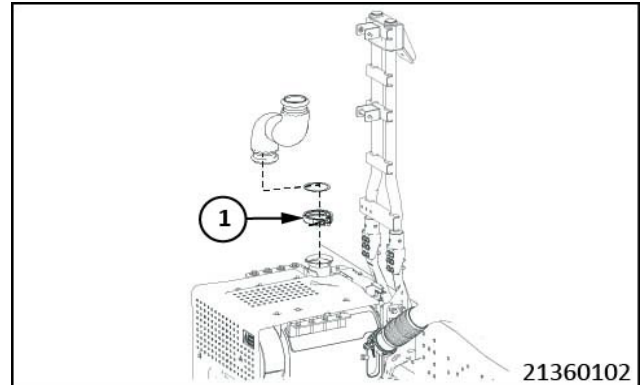
3. Torque the vertical exhaust bracket bolt (1) and nut (2) between 20 to 25 Nm using a torque wrench with a 13 mm socket.



4. Place the center pipe exhaust (1) and install the V-band clamp (2) with gasket (3).
5. Tighten the clamp (2) using a V-band clamp tool.

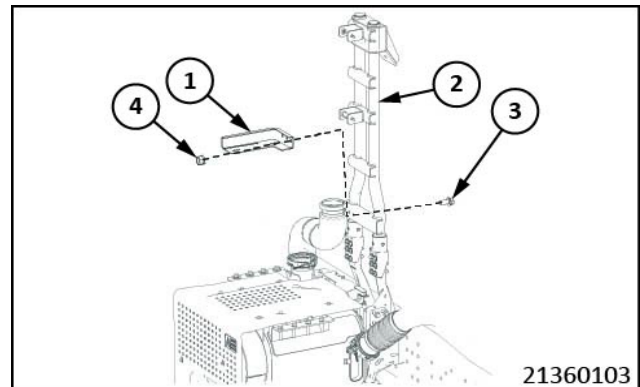


6. Torque the V-band clamp(1) between 5 to 7 Nm using a torque wrench.

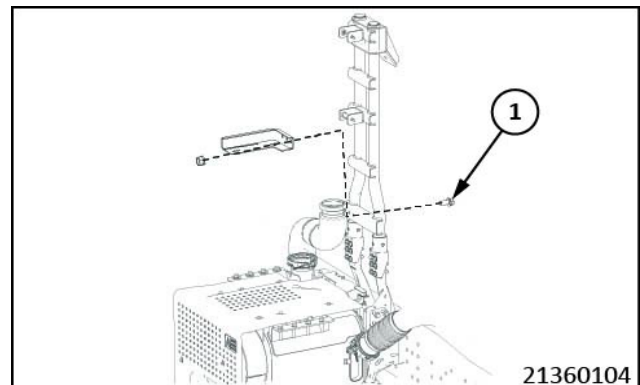


7. Attach the exhaust bracket (1) to the vertical exhaust (2) using the bracket bolt (3) and nut (4).

8. Tighten the bolt (3) using a 13 mm socket and ratchet.

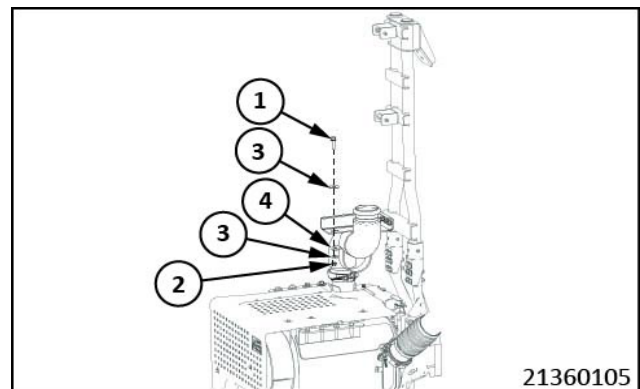


9. Torque the exhaust bracket bolt (1) between 20 to 25 Nm using a torque wrench with a 13 mm socket.

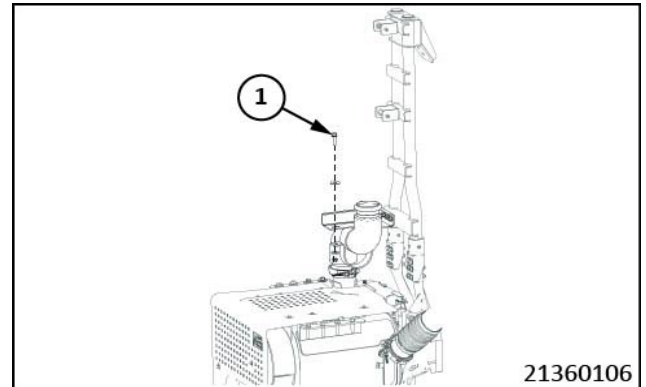


10. Place the bolt (1) and nut (2) along with washer (3) into the centre pipe exhaust clamp (4).

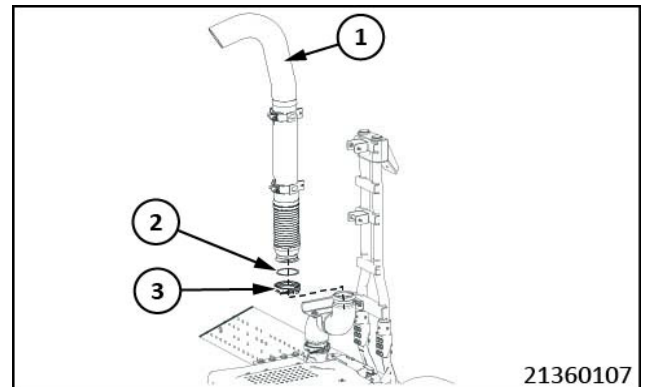
11. Tighten the center pipe clamp mounting bolt (1) using a 17 mm socket and ratchet.



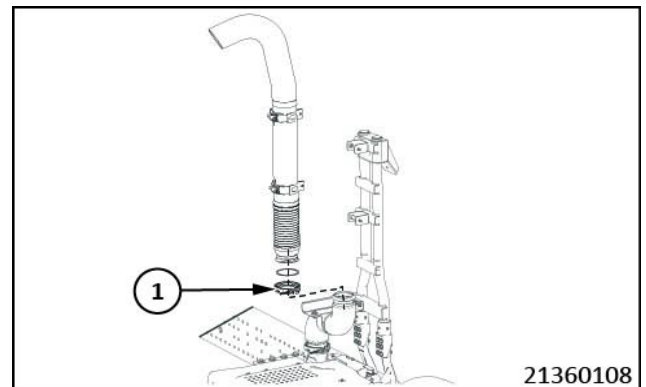
12. Torque the center pipe clamp mounting bolt (1) between 22 to 25 Nm using a torque wrench with a 17 mm socket.



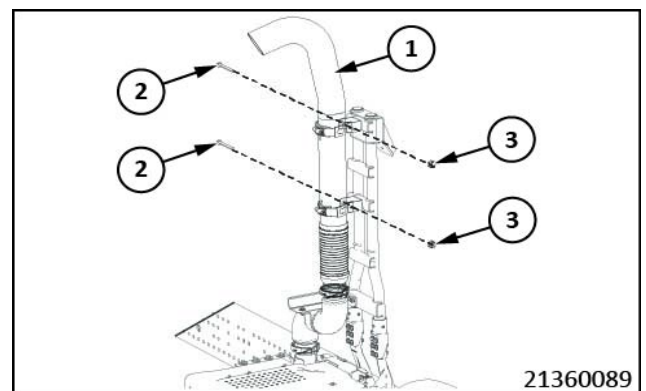
13. Position the vertical exhaust tail pipe (1) and insert the gasket (2). Install the V-band clamp (3) and tighten using a V-band clamp tool.



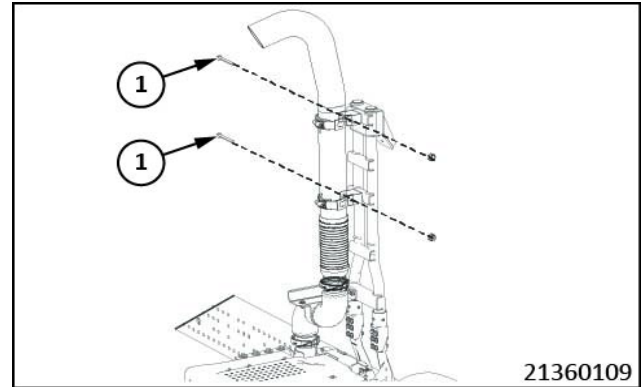
14. Torque the V-band clamp (1) between 5 to 7 Nm using a torque wrench.



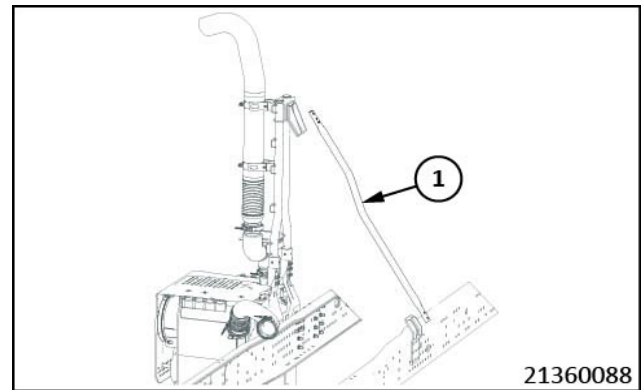
15. Install the tail pipe vertical mounting clamp (1) bolt (2) and nut (3). Tighten using a 13 mm socket and ratchet.



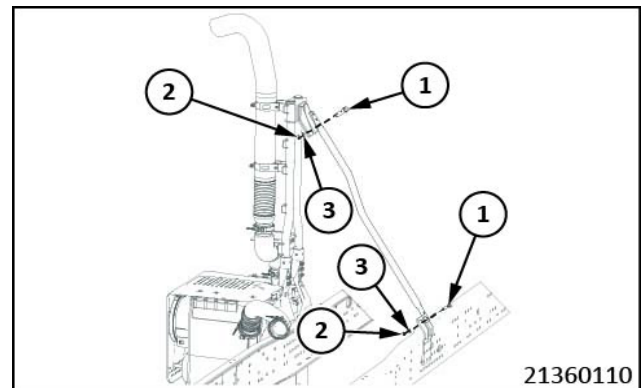
16. Torque the clamp bolt (1) between 8 to 11 Nm using a torque wrench with a 13 mm socket.



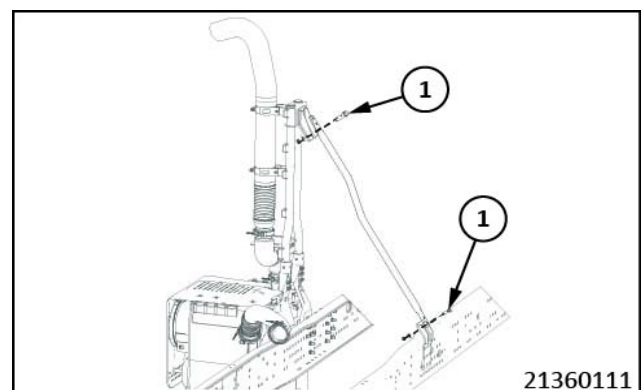
17. Refit the vertical exhaust stay (1) and align it to its mounting points.



18. Install the upper and lower mounting bolts (1) and nuts (2) along with washers (3). Tighten the bolts using a 13 mm socket and ratchet.



19. Torque both bolts (1) between 20 to 25 Nm using a torque wrench with a 13 mm socket.



36.1.6.10 Alignment Check

1. Ensure the tail pipe is aligned and not touching any other vehicle component.

ADD
IMAGE

2. All clamps must be seated correctly and tightened to prevent exhaust leakage.

ADD
IMAGE

3. Verify clearance between the tail pipe and the bodywork.

ADD
IMAGE

36.1.6.11 Final Testing

1. Start the engine and check for any abnormal noise or leakage.

ADD
IMAGE

2. Observe vibration or alignment issues visually.

ADD
IMAGE

3. Perform a short idle and road test.

ADD
IMAGE

4. Recheck torque settings if needed.

ADD
IMAGE

36.1.6.12 Warranty Clause

This activity falls under the following applicable warranties:

- Standard Vehicle Warranty
- Parts Warranty
- Labour Warranty
- Aggregate Warranty
- Consumables Warranty
- Extended / Driveline Warranty
- LCP Scheme (Labour, Consumables, Parts)
- AMC (Annual Maintenance Contract)

- Uptime / Eicher Promise
- Goodwill Warranty (if approved)